

Chemistry — Part 1

Some Basic Concepts of Chemistry · Atomic Structure

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SOME BASIC CONCEPTS OF CHEMISTRY

DAY 01

- Life saving drugs like taxol and cisplatin. (used in cancer therapy)
- Azidothymidine (AZT) used for AIDS victims.

NATURE OF MATTER

- Matter is anything that occupies space, possesses mass, offers resistance and can be felt by one or more of our senses.
- Examples : water, air, metals, plants, animals, etc.
- The matter can be classified into different categories depending upon its physical or chemical nature.
- Matter can mainly exist in three physical states viz. solid, liquid and gas.

[Figure] Particle arrangement diagrams for solid, liquid and gas

- The constituents are very closely packed in solid and have less freedom of movement.
- In liquids, the particles are close to each other but can move around.
- In gases, the constituents are far apart and can move freely and easily.
- The three states of matter can be interconverted into one another by changing temperature and pressure conditions.

Solid $\xrightarrow[\text{cool}]{\text{heat}}$ liquid $\xrightarrow[\text{cool}]{\text{heat}}$ Gas

[Figure] Circular flowchart of interconversion between solid, liquid and gas
(melting/freezing/evaporation/condensation/sublimation/deposition)

CHEMICAL CLASSIFICATION OF MATTER

SOLIDS	LIQUIDS	GASES
Have a definite shape and volume.	Have a definite volume but no definite shape.	Neither have a definite shape nor volume.
Intermolecular force of attraction is very high	Intermolecular force of attraction is moderate	Intermolecular force of attraction is very low
Have melting point above the room temperature.	The melting point is below the room temperature.	The boiling point is below the room temperature.

MATTER — PURE SUBSTANCES | — ELEMENTS (Eg : C, O₂) | — COMPOUNDS (Eg : CO₂, NH₃)
 — MIXTURES — HOMOGENEOUS (Eg : Air, alloys) — HETEROGENEOUS (Eg : Dust in Air, Sand)

- Based upon chemical composition of various substances matter can be classified into two major categories, pure substances and mixtures.
- Pure substances can be further divided into two types, elements and compounds.
- Mixtures are also of two types, homogeneous mixtures and heterogeneous mixtures.

ELEMENTS

- An element may be defined as a pure substance which can neither be decomposed into nor built from simpler substances by any physical or chemical method.
- An element is pure substance which is made of one kind of atoms.
- There are about 118 elements known so far. Out of these elements 92 occur in nature.

COMPOUNDS

- A compound may be defined as a pure substance which can be decomposed into simpler substances by some suitable chemical method.
- A compound is formed by combination of two or more elements in a definite proportion by mass.
- When a compound is formed, the constituent elements lose their properties.
- For example, hydrogen is combustible, oxygen is supporter of combustion but water which is a compound of hydrogen and oxygen is neither combustible nor supporter of combustion rather it puts off fire.

MIXTURES

- Mixtures may be defined as the substances which are made up of two or more pure substances.
- Mixtures are of two types, homogeneous mixtures and heterogeneous mixtures.

01. HOMOGENEOUS MIXTURES

- They are the mixtures which have same composition throughout.
- These mixtures are also known as solutions.
- Some examples of homogeneous mixtures are air, gasoline, sea water, brass, coloured glass, etc.

02. HETEROGENEOUS MIXTURES

- They are the mixtures which have different composition in different parts.
-

- In other words, components of such a mixture are not distributed uniformly.
- The components of a heterogeneous mixture can be observed with naked eyes or with the help of a microscope.
- Some examples of heterogeneous mixtures are sand and iron filings, sand and water, mud, etc.

[Figure]

Particle arrangement diagrams showing homogeneous (uniformly distributed particles) and heterogeneous (non-uniformly distributed particles) mixtures

BOARD QUESTION

1. Classify the following into elements and compounds ?

O₂, CO₂, H₂O, H₂SO₄, CH₄, Cl₂, H₂, Au

Answer : Elements : O₂, Cl₂, H₂, Au Compounds : CO₂, H₂O, H₂SO₄, CH₄

BOARD QUESTION

1. Classify the following into Homogeneous and Heterogeneous mixtures ?

Air, sand and water, gasoline, sea water, sand and iron filings, brass, coloured glass, mud.

Answer : Homogeneous mixtures : Air, gasoline, sea water, brass, coloured glass. Heterogeneous mixtures : Sand and water, sand and iron filings, mud.

TRY YOURSELF

1. What is matter and how is it classified?

TRY YOURSELF

1. What are homogeneous and heterogeneous mixtures. Explain with examples?

TRY YOURSELF

1. Define the following

i) Element ii) Compound

DAY 02

PROPERTIES OF MATTER AND THEIR MEASUREMENT

- Every substance has unique or characteristic properties. These properties can be classified into two categories,
- Physical Properties
- Chemical Properties
- Physical properties are those properties which can be measured or observed without changing the identity or the composition of the substance.
- Examples : colour, odour, melting point, boiling point, density, etc.
- Chemical properties are those properties whose measurement or observation requires a chemical change to occur.
- Examples : Acidity or basicity, combustibility, etc.

INTERNATIONAL SYSTEM OF UNITS (SI)

- In 1960, the International Committee of Weights and Measures recommended the use of International System of Units, abbreviated as SI Units.
- The seven basic units in SI system are given in the following table.

Physical Quantity	Abbreviation	Name of Unit	Symbol
Length	l	metre	m
Mass	m	kilogram	kg
Time	t	second	s
Electric current	I	ampere	A
Thermodynamic Temperature	T	kelvin	K
Luminous intensity	I_v	candela	cd
Amount of substance	n	mole	mol

MEASUREMENT

- The quantities which we come across during the scientific studies are called physical quantities.
- A unit may be defined as the standard of reference chosen to measure or express any physical quantity.

[Figure] Two images: left shows a glass bell jar with a metallic object inside, right shows a golden metallic weight with a circular emblem

- The other physical quantities, such as speed, volume, density, etc., can be derived from these quantities.
- The fundamental units of metric system are gram for mass, the metre for length and the litre for the volume of fluids.

MASS AND WEIGHT

- Mass of a substance is the amount of matter present in it while weight is the force exerted by gravity on an object.
 - The mass of a substance is constant whereas its weight may vary from one place to another due to change in gravity.
-

VOLUME

VOLUME

- Volume is the amount of space occupied by a substance.
- Volume has the units of (length)³.
- SI system, volume has units of m³.
- Smaller units such as cm³ or dm³ are also used.
- A common unit, litre (L) which is not an SI unit, is used for the measurement of volume of liquids

$$1\text{m}^3 = 10^3\text{L} = 10^3\text{dm}^3 = 10^6\text{cm}^3 = 10^6\text{mL} \quad 1\text{L} = 1\text{dm}^3 \quad 1\text{mL} = 1\text{cm}^3$$

DENSITY

DENSITY

- Density of a substance is its amount of mass per unit volume.
- SI units of density can be obtained as follows:

$$\text{SI unit of density} = \frac{\text{SI unit of mass}}{\text{SI unit of volume}} = \text{kg} / \text{m}^3$$

- It can be expressed in g / cm³ also, where mass is expressed in gram and volume is expressed in cm³.

TEMPERATURE

TEMPERATURE

- Temperature is the degree of hotness or coldness
- There are three common scales to measure temperature

Degree celsius : °C Degree Fahrenheit : °F Kelvin : K

- Kelvin (K) is the SI unit.

$$T(^{\circ}\text{F}) = \frac{9}{5} T(^{\circ}\text{C}) + 32, T(\text{K}) = T(^{\circ}\text{C}) + 273.15$$

PREFIXES FOR EXPRESSING THE DECIMAL FRACTIONS IN THE SI SYSTEM

MULTIPLE	PREFIX	SYMBOL
10 ²⁴	yotta	Y
10 ²¹	zetta	Z
10 ¹⁸	exa	E
10 ¹⁵	peta	P
10 ¹²	tera	T

MULTIPLE	PREFIX	SYMBOL
10^9	giga	G
10^6	mega	M
10^3	kilo	k
10^2	hecto	h
10	deca	da
10^{-1}	deci	d
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p
10^{-15}	femto	f
10^{-18}	atto	a
10^{-21}	zepto	z
10^{-24}	yocto	y

BOARD QUESTION

01. Fill in the blanks.

Physical quantity	SI Unit
Mass	-----
-----	Kelvin
-----	Candela
Pressure	-----

Answer :

Physical quantity	SI Unit
Mass	Kilogram
Temperature	Kelvin
Luminous intensity	Candela
Pressure	Pascal

BOARD QUESTION

02. Match the following prefixes with their multiples:

	Prefixes	Multiples
(i)	micro	10^6
(ii)	deca	10^{-6}
(iii)	mega	10^9
(iv)	giga	10^{-15}
(v)	femto	10

Answer : Micro : 10^{-6} Deca : 10 Mega : 10^6 Giga : 10^9 Femto : 10^{-15}

TRY YOURSELF

04. What do you mean by temperature ? Write its different units. How are they inter related?

TRY YOURSELF

05. What do you mean by volume ? Write its SI units. write the relation between SI unit of volume with litre ?

DAY 03

SCIENTIFIC NOTATION

- In scientific notation or exponential notation any number can be represented in the form $N \times 10^n$, where n is an integer exponent having zero, positive or negative values and N can vary between 1 to 10, i.e, $1 \leq N < 10$.
- For example, we can write 136.225 as 1.36225×10^2 in scientific notation.

SIGNIFICANT FIGURES

- A convenient method of expressing the uncertainty in measurement is to express it in terms of significant figures.
- The number of significant figures in a measurement is the number of figures that are known with certainty plus one that is uncertain, beginning with the first non-zero digit.

In order to determine the significant figures in a measured quantity the following rules should be applied,

- All non-zero digits are significant.
- 165 cm has three significant figures.
- Zeros to the left of the first non-zero digit in the number are not significant.
- 0.026 g has two significant figures.
- Zeros between non-zero digits are significant.
- 0.4006 g has four significant figures.
- Zeros to the right of the decimal point are significant.
- 5.00 g, 0.050 g, 0.5000 g have three, two and four significant figures respectively.
- If a number ends in zeros that are not to the right of a decimal, the zeros may or may not be significant.
- 2180 has three significant figures.
- In scientific notation the number is written in the standard exponential form as $N \times 10^n$.
- 1.5×10^3 g (Two significant figures)
- 1.50×10^3 g (Three significant figures)
- The exponential notation is an excellent way of expressing significant figures in very large or very small measurements.
- Avogadro's constant, $6.022 \times 10^{23} \text{ mol}^{-1}$: Four significant figures
- Planck's constant, $6.625 \times 10^{-34} \text{ Js}$: Four significant figures
- Lastly, it may be emphasized that exact integral numbers, such as number of pencils in a dozen pencils or number of grams in a kilogram do not have any uncertainty associated with them and as such these numbers have an infinite number of significant figures.
- The value of π has infinite number of significant figures.

ROUNDING OFF

The rules to be followed are :

- If the digit following the last digit to be retained is less than five, the last digit is left unchanged.

- If the digit following the last digit to be retained is greater than five, the last digit retained is increased by one.
 - If the digit following the last digit to be retained is equal to five, the last digit is left unchanged if it is even and is increased by one if it is odd.
 - In addition and subtraction, the final result should be reported to the same number of decimal places as the number with the fewest number of decimal places.
 - In multiplication and division, the final result should be reported as having the same number of significant digits as the number with fewest number of significant digits.
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LAWS OF CHEMICAL COMBINATIONS

01. LAW OF CONSERVATION OF MASS

- By studying the results of quantitative measurements of many reactions it was observed that whenever substances react, they do so according to certain laws. These laws are called the laws of chemical combinations.
- This law was proposed by French chemist Antoine Laurent Lavoisier (1789) and verified by Landolt.
- Law of conservation of mass is also known as law of indestructibility of matter.
- This law states that: During any physical or chemical change, the total mass of the products remains equal to the total mass of the reactants.
- Or matter can neither be created nor be destroyed during an ordinary chemical reaction

KEEP IT IN MIND - Law of conservation of mass is not applicable in the case of nuclear reactions. - Balancing of a chemical equation is based on Law of conservation of mass.

BOARD QUESTION

1. Express the following in the scientific notation:

i) 0.0048 ii) 234000 iii) 8008 iv) 500.0

Answer : i) 4.8×10^{-3} ii) 2.34×10^5 iii) 8.008×10^3 iv) 5.000×10^2

BOARD QUESTION

1. How many significant figures are present in the following?

i) 0.0025 ii) 208 iii) 500.0 iv) 2.0034

Answer : i) 2 ii) 3 iii) 4 iv) 5

BOARD QUESTION

1. How many significant figures should be present in the answer of the following calculations?

i) $(0.02856 \times 298.15 \times 0.112) / 0.5785$ ii) 5×5.364 iii) $0.0125 + 0.7864 + 0.0215$

Answer : i) $(0.02856 \times 298.15 \times 0.112) / 0.5785 = 1.64$ 3 significant figures, as the least precise term i.e., 0.112 has 3 significant figures. ii) $5 \times 5.364 = 26.82$, 4 significant figures, as 5 is exact number but the other term, i.e., 5.364 has 4 significant figures. iii) $0.0125 + 0.7864 + 0.0215 = 0.8204$, i.e., 4 significant figures as the least number of decimal place is 4.

BOARD QUESTION

1. 10.0 g of CaCO_3 on heating gave 4.4 g CO_2 and 5.6 g CaO . Show that these observations are in agreement with law of conservation of mass.

Answer : Mass of the reactant = 10.0 g Mass of the products = 4.4 + 5.6 = 10.0 g Since mass of the reactants is equal to the mass of the products, therefore, these observations are in agreement with Law of conservation of mass.

02. LAW OF CONSTANT COMPOSITION OR DEFINITE PROPORTIONS

• This law deals with the composition of chemical compounds. • It was discovered by the French chemist, Joseph Proust (1799) and verified by Stas and Richards. • This law states that a chemical compound always contains same elements combined together in same proportion by mass. • This law implies that in a chemical compound the elements are present in fixed and not arbitrary ratio by mass. • For example, pure water obtained from different sources such as, rivers, wells, springs, seas, etc., always contains hydrogen and oxygen combined in the ratio 1: 8 by mass. • This law is not applicable in the case of non-stoichiometric compounds and when same compound is formed using different isotopes of the same element.

The mass ratio of A to B is the same in both samples, illustrating the law of constant composition.

TRY YOURSELF

TRY YOURSELF

1. How many significant figures are present in
 $0.01627 \times 321.45 \times 0.874 \times 0.9283$

TRY YOURSELF

1. Express the following in the scientific notation:
 i) 0.0028 ii) 78900 iii) 2002 iv) 20.0 v) 2.10 vi) 0.00089

BOARD QUESTION

1. In one sample of a compound of A and B, 2.50g A combines with 8.00 g B. In another sample of a compound of A and B, 1.20 g of A combines with 3.84 g of B. Show that these results illustrate law of constant composition?

Answer : The mass ratio of A to B in sample 1 $\frac{2.50 \text{ g}}{8.00 \text{ g}} = 0.3125$

The mass ratio of A to B in sample 2 $\frac{1.20 \text{ g}}{3.84 \text{ g}} = 0.3125$

DAY 04

03. LAW OF MULTIPLE PROPORTIONS

- This law was discovered by John Dalton (1803).
- This law states that: When two elements combine with each other to form two or more compounds, the masses of one of the elements which combine with fixed mass of the other, bear a simple whole number ratio to one another.
- Not applicable when same compound is obtained by using different isotopes of the same element

BOARD QUESTION 01. Hydrogen and oxygen are known to form two compounds. The hydrogen content in one of these is 5.93% while in the other it is 11.2%. Show that this data illustrates the Law of multiple proportions?

Answer : In the first compound Hydrogen = 5.93% Oxygen = $(100 - 5.93)\% = 94.07\%$. In the second compound Hydrogen = 11.2% Oxygen = $(100 - 11.2)\% = 88.8\%$ In the first compound the number of parts by mass of oxygen that combine with one part by mass of hydrogen = $94.07/5.93 = 15.86$ parts. In the second compound the number of parts by mass of oxygen that combine with one part by mass of hydrogen = $88.8/11.2 = 7.93$ parts. The ratio of masses of oxygen that combine with fixed mass (1 part) by mass of hydrogen is $15.86 : 7.9$ or $2 : 1$. Since this ratio is a simple whole number ratio, hence the given data illustrates the law of multiple proportions.

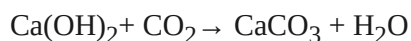
TRY YOURSELF

1. Round off the following upto three significant figures:

i) 0.0028 ii) 78900 iii) 20026 iv) 0.00089

TRY YOURSELF

1. Does the following chemical reaction obey the law of conservation of mass?



04. GAY-LUSSAC'S LAW OF COMBINING VOLUMES

- This law was proposed by Gay Lussac.
- This law states that: When gases react with each other they do so in volumes that bear simple whole-number ratio to one another and to the volumes of gaseous products, provided all volumes are measured under similar conditions of temperature and pressure.

05. AVOGADRO'S LAW

- Avogadro's Law, states that: Equal volumes of all gases, under similar conditions of temperature and pressure, contain equal numbers of molecules.

KEEP IT IN MIND

- If the molecule is made of atoms of the same element, it is called a homoatomic molecule.
- They may be monoatomic, diatomic or polyatomic depending on whether as they contain, one, two or more than two atoms, respectively.
- Some common examples of homoatomic molecules are He, N₂, O₂, O₃, S₈, etc.
- The number of atoms present in one molecule of element is called atomicity of the element.
- If the molecule is made of atoms of different elements, it is called a heteroatomic molecule.
- They may be diatomic or polyatomic depending upon the number of atoms they contain.
- Some examples of heteroatomic molecules are CO, CO₂, H₂O, SO₃, H₂SO₄, etc.

DALTON'S ATOMIC THEORY

By observing the laws of chemical combination, John Dalton (1808) proposed the atomic theory of matter.

- The main points of Dalton's atomic theory are as follows:
- Matter is made up of extremely small, indivisible particles called atoms.
- Atoms of same element are identical in all respects, i.e., they possess same size, shape, mass, chemical properties, etc.
- Atoms of different element are different in all respects, i.e., they possess different sizes, shapes, masses, chemical properties, etc.
- An atom is the smallest particle that takes part in chemical reactions.
- Atoms of different elements may combine with each other in a fixed, simple, whole number ratio to form compound atoms.
- Atoms can neither be created nor destroyed, i.e., atoms are indestructible.

ATOMIC MASS

- The scale in which the relative atomic masses of different atoms are expressed is called atomic mass unit scale or a.m.u. scale.
- The new symbol 'u' (unified atomic mass unit) is recommended to be used in place of amu. The atomic-mass unit or unified mass may be defined as one-twelfth of the mass of an atom of a carbon-12 isotope.

- The atomic mass of an element tells us as to how many times an atom of the element is heavier than one-twelfth of an atom of carbon-12 isotope.

Atomic mass of an element = Mass of an atom of the element $\frac{1}{12}$ the mass of an atom of carbon(C-12)

GRAM ATOMIC MASS (GAM)

- It may be defined as that much quantity of the element whose mass in gram is numerically equal to its atomic mass.
- Gram atomic mass is also called one gram-atom of the element.

Number of g atoms = Mass in gram / Gram atomic mass

Element	Mass of an atom	Gram Atomic Mass
H	1 amu	1 g
He	4 amu	4 g
Li	7 amu	7 g
Be	9 amu	9 g
B	11 amu	11 g
C	12 amu	12 g
N	14 amu	14 g
O	16 amu	16 g
F	19 amu	19 g
Ne	20 amu	20 g
Na	23 amu	23 g
Mg	24 amu	24 g
Al	27 amu	27 g
Si	28 amu	28 g
P	31 amu	31 g
S	32 amu	32 g
Cl	35.5 amu	35.5 g
Ar	40 amu	40 g
K	39 amu	39 g
Ca	40 amu	40 g
Sc	45 amu	45 g
Ti	48 amu	48 g
V	51 amu	51 g

Sc	45 amu	45 g
Cr	52 amu	52 g
Mn	55 amu	55 g
Fe	56 amu	56 g
Co	59 amu	59 g
Ni	58.7 amu	58.7 g
Cu	63.5 amu	63.5 g
Zn	65.4 amu	65.4 g
Br	80 amu	80 g
I	127 amu	127 g
Ag	108 amu	108 g
Ba	137 amu	137 g
Au	197 amu	197 g

AVERAGE ATOMIC MASS

- In the case of elements having isotopes, atoms of the element possess different relative atomic masses.
- In such cases atomic mass of the element is the weighted average of relative atomic masses of its naturally occurring isotopes.

$$\text{Average atomic mass} = (M_1X + M_2Y) / (X + Y)$$

where M_1 and M_2 are isotopic masses. X and Y are percentage abundance or fractional abundance.

- If X and Y are percentage abundance, then $X + Y = 100$
 - If X and Y are fractional abundance, then $X + Y = 1$
-

MOLECULAR MASS

MOLECULAR MASS

- Molecular mass of a substance tells us the number of times a molecule of the substance is heavier than 1/12th of mass of a carbon (C-12) atom.
- Molecular mass of water (H₂O) is 18 u. It means that a molecule of water is 18 times heavier than 1/12th of mass of a carbon atom (C-12).

MOLAR MASS (M)

- It may be defined as that much quantity of the substance (element or compound) whose mass in gram is numerically equal to its molecular mass.
- Molar mass is also called one gram-molecule of the substance.

Number of g molecules = Mass in gram / Gram molecular mass

1. Calculate the atomic mass (average) of chlorine using the following data:

	% Natural Abundance	Isotopic Molar Mass
³⁵ Cl	75.77	34.9689
³⁷ Cl	24.23	36.9659

Answer :

Given, % natural abundance of ³⁵Cl = 75.77 Molar mass of ³⁵Cl = 34.9689 % natural abundance of ³⁷Cl = 24.23 Molar mass of ³⁷Cl = 36.9659

Average atomic mass = $(75.77 \times 34.9689 + 24.23 \times 36.9659) / (75.77 + 24.23) = 35.4527$ amu

1. Nitrogen occurs in nature in the form of two isotopes with atomic mass 14 and 15 respectively.

If average atomic mass of nitrogen is 14.0067, what is the % abundance of the two isotopes?

Answer :

Let % abundance of N-14 isotope = x Then % abundance of N-15 isotope = (100-x)

The average atomic mass = $(x \times 14 + (100-x) \times 15) / 100$

Now, $(14x + (100-x) \times 15) / 100 = 14.0067$ $-x = 1400.67 - 1500 = -99.33$ or $x = 99.33$

Thus, % abundance of N-14 isotope = 99.33% abundance of N-15 isotope = 0.67%.

Space for Keynotes

DAY 05

MOLE CONCEPT

Compound	Mass of a molecule	Molar mass/gram molecular mass
H ₂	2 amu	2 g
N ₂	28 amu	28 g
O ₂	32 amu	32 g
Cl ₂	71 amu	71 g
HCl	36.5 amu	36.5 g
HNO ₃	63 amu	63 g
H ₂ SO ₄	98 amu	98 g
H ₂ CO ₃	62 amu	62 g
H ₃ PO ₄	98 amu	98 g
NaOH	40 amu	40 g
KOH	56 amu	56 g
Ca(OH) ₂	74 amu	74 g
NaCl	58.5 amu	58.5 g
AgCl	143.5 amu	143.5 g
AgBr	188 amu	188 g
AgI	235 amu	235 g
BaSO ₄	233 amu	233 g
BaCl ₂	208 amu	208 g
NH ₂ CONH ₂	60 amu	60 g
CH ₃ COOH	60 amu	60 g
C ₆ H ₁₂ O ₆	180 amu	180 g
C ₁₂ H ₂₂ O ₁₁	342 amu	342 g

The word mole was introduced by Wilhelm Ostwald who derived the term from the Latin word moles meaning a heap or pile.

The mole may be defined as “the amount of the substance that contains as many specified elementary particles as the number of atoms in 12 g of carbon-12 isotope”.

This number has been experimentally found to be 6.022×10^{23} and is called one mole.

This number is also known as Avogadro’s number or Avogadro constant ($N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$).

ONE MOLE = 6.022×10^{23} PARTICLES

[Figure] Diagram showing two weighing scales with bowls of particles labeled "Carbon atoms" and "Aluminium atoms", with text " 6.022×10^{23} atoms" above and "1 mole of C=12g" and "1 mole of Al=27g" below

KEEP IT IN MIND - One mole of Oxygen atoms $\approx 6.023 \times 10^{23}$ oxygen atoms - One mole of Oxygen molecules $\approx 6.023 \times 10^{23}$ oxygen molecules - One mole of NH_3 molecule $\approx 6.023 \times 10^{23}$ ammonia molecules - One mole of Chloride ions $\approx 6.023 \times 10^{23}$ chloride ions

MOLAR VOLUME

KEEP IT IN MIND

- 1 mole $C_{12}H_{22}O_{11}$ contains
- 1 mole $C_{12}H_{22}O_{11}$ molecules
- 6.022×10^{23} $C_{12}H_{22}O_{11}$ molecules
- 12 mol C atoms
- $12 \times 6.022 \times 10^{23}$ C atoms
- 22 mol H atoms
- $22 \times 6.022 \times 10^{23}$ H atoms
- 11 mol O atoms
- $11 \times 6.022 \times 10^{23}$ O atoms
- 45 mol of total atom
- $45 \times 6.022 \times 10^{23}$ atoms
- n mole $C_{12}H_{22}O_{11}$ contains
- n mole $C_{12}H_{22}O_{11}$ molecules
- $n \times 6.022 \times 10^{23}$ $C_{12}H_{22}O_{11}$ molecules
- $n \times 12$ mol C atoms
- $n \times 12 \times 6.022 \times 10^{23}$ C atoms
- $n \times 22$ mol H atoms
- $n \times 22 \times 6.022 \times 10^{23}$ H atoms
- $n \times 11$ mol O atoms
- $n \times 11 \times 6.022 \times 10^{23}$ O atoms

BOARD QUESTION

1. 0.5 mole H_2SO_4 contains

MOLAR VOLUME

- The volume occupied by one mole molecules of a gaseous substance is called Molar Volume or Gram Molecular Volume (G.M.V.).
- One mole molecules of all gases occupy 22.4 litres at 273 K and 760 mm pressure (S.T.P.).

[Figure] Three balloons each labeled "22.4 L" attached to gas cylinders, showing 1 mol of different gases at S.T.P.

Condition	Temp.	Press.	Molar volume
Old S.T.P	273.15 K	1 atm	$22.4 \text{ L} = 22.4 \times 10^{-3} \text{ m}^3$

Condition	Temp.	Press.	Molar volume
New S.T.P	273.15 K	1 bar	22.7 L = $22.7 \times 10^{-3} \text{m}^3$
S.A.T.P	298.15 K	1 bar	24.8 L = $24.8 \times 10^{-3} \text{m}^3$

- The previous standard (old S.T.P) is still often used, and applies to all chemistry data.
- $22.4 \text{ L} = 22400 \text{ mL}$

$$= 22400 \text{ cm}^3 = 22.4 \text{ dm}^3 = 22.4 \times 10^{-3} \text{m}^3$$

1 mole = 1 g atom (for elements) = 1 g molecule (for compounds) = 6.022×10^{23} particles = 22.4 L (for any gas at S.T.P)

KEEP IT IN MIND

No. of moles = No. of gram atom = Mass in gram / Gram atomic mass
 No. of moles = No. of gram molecules = Mass in gram / Gram molecular mass
 No. of moles = Volume in L / 22.4 L
 No. of moles = Volume in mL / 22400 mL
 No. of moles = No. of particles / Avogadro number

- Mass of one mole of atoms = gram atomic mass
- Mass of one mole of molecules = Molar mass
- Mass of 22.4 L of any gas at S.T.P = Molar mass

BOARD QUESTION

1. Calculate the mass of a silver atom: (atomic mass of silver = 108 g)

Answer : Mass of 6.022×10^{23} atoms of Ag = 108 g
 Mass of one atom of Silver $108 / 6.022 \times 10^{23} = 1.79 \times 10^{-22}$ g

BOARD QUESTION

1. Calculate the mass of one molecule of water

Answer : 6×10^{23} molecules has mass = 18 g
 1 molecule has mass = $18 / (6 \times 10^{23}) = 3 \times 10^{-23}$ g = 3×10^{-26} kg

BOARD QUESTION

1. Calculate the number of molecules present in 1 g water:

Answer : Number of moles in 1 g water = $1/18$ $\therefore$ No. of molecules in 1 g water = $1 \times 6.022 \times 10^{23} = 3.35 \times 10^{22}$

BOARD QUESTION

1. Calculate the number of atoms present in 25 g of carbon:

Answer : We know that 12 g of carbon contains 6×10^{23} atoms (1 mole of carbon = 12 g) Avogadro number $\approx 6 \times 10^{23}$
 25 g of carbon contains = $25 \times 6 \times 10^{23} / 12 = 12.5 \times 10^{23}$ atoms

BOARD QUESTION

1. How many oxygen atoms are present in 50 g of CaCO_3 ?

Answer : Molecular mass of $\text{CaCO}_3 = 40 + 12 + 3 \times 16 = 100$ g
 Atoms of oxygen present in 100 g of $\text{CaCO}_3 = 3 \times 6.022 \times 10^{23}$ atoms.
 Atoms of oxygen present in 50 g of $\text{CaCO}_3 = \{(3 \times 6.022 \times 10^{23}) \times 50 / 100\}$ atoms = 9.033×10^{23}

BOARD QUESTION

1. Calculate the number of molecules present in 56 mL of CO_2 at STP:

Answer : Volume = 56 mL = 0.056 L

Molar volume of any gas at STP = 22.4L Number of moles = volume/(22.4 L) Number of moles = $0.056/22.4 = 0.0025$ L No.of molecules = $0.0025 \times 6.022 \times 10^{23} = 0.01505 \times 10^{23}$

= $[(2 \times 23.0) + (32.00) + 4(16.00)] = 142$ g Mass percent of an element = (Mass of that element in compound/Molar mass of that compound) $\times 100$ $\therefore$ Mass percent of sodium (Na): $(46/142) \times 100 = 32.39\%$ Mass percent of sulphur (S): $(32/142) \times 100 = 22.54\%$ Mass percent of oxygen: (O): $(64/142) \times 100 = 45.07\%$

PERCENTAGE COMPOSITION

- Percentage composition of a compound

refers to the amount of various constituent elements present per 100 parts by mass of the substance.

- It can be calculated from the formula of the

compound by using the following relation: % (mass) of an element = (No.of atoms of element) $\times$ (At.mass of element) $\div$ Formula mass of substance $\times 100$

- In case of minerals or ores the mass

percentage of a particular constituent species in a formula can also be calculated using the similar relation: % (mass) of constituent species = [Total mass of constituent species in one formula unit] $\div$ Formula mass $\times 100$

BOARD QUESTION

1. Calculate the mass percent of different elements present in sodium sulphate (Na_2SO_4). Answer : Molar mass of Na_2SO_4

TRY YOURSELF

1. Calculate the mass of a calcium atom:

TRY YOURSELF

1. Calculate the mass of a SO_2 molecule:

DAY 06

DETERMINATION OF MOLECULAR FORMULA

- The molecular formula of a compound may be defined as the formula which gives the actual number of atoms of various elements present in the molecule of the compound.
- The empirical formula of a compound may be defined as the formula which gives the simplest whole number ratio of atoms of the various elements present in the molecule of the compound.
- The empirical formula of the compound glucose ($C_6H_{12}O_6$), is CH_2O which shows that C, H and O are present in the simplest ratio of 1 : 2 : 1.

KEEP IT IN MIND - Empirical formula mass of substance is equal to the sum of atomic masses of all the atoms in the empirical formula of the substance. - Molecular formula is whole number multiple of empirical formula. - $\text{Molecular formula} = \text{Empirical formula} \times n$ $n = \text{Molecular formula mass} / \text{Empirical formula mass}$

STEPS FOR WRITING THE EMPIRICAL FORMULA

- The empirical formula is determined by the following steps:
 - > Divide the percentage of each element by its atomic mass. This will give the relative number of moles of various elements present in the compound.
 - > Divide the quotients obtained in the above step by the smallest of them so as to get a
 - TRY YOURSELF
 - 1. How many atoms are present in 50 g of copper?
 - TRY YOURSELF
 - 1. Calculate the number of atoms present in 3.4 g of hydrogen sulphide (H_2S) :
 - TRY YOURSELF
 - 1. Calculate the number of atoms present in 56 mL of O_2 at STP:
-

simple ratio of moles of various elements.

- Multiply the figures, so obtained by a suitable integer if necessary in order to obtain whole number ratio.
- Finally, write down the symbols of the various elements side by side and put the above numbers as the subscripts to the lower right hand corner of each symbol. This will represent the empirical formula of the compound.

iii) HO iv) H_2O v) Na_2CO_3 vi) BH_3 vii) NO_2 viii) H_3PO_4 ix) Fe_2O_3 x) CH

STEPS FOR WRITING THE MOLECULAR FORMULA

- Calculate the empirical formula as described above.
- Find out the empirical formula mass by adding the atomic masses of all the atoms present in the empirical formula of the compound.
- Divide the molecular mass (determined experimentally by some suitable method) by the empirical formula mass and find out the value of n .
- Multiply the empirical formula of the compound with n so as to find out the molecular formula of the compound.

BOARD QUESTION

1. Write the empirical formula of the compounds having molecular formulae:

i) C_6H_6 ii) C_6H_{12} iii) H_2O_2 iv) H_2O v) Na_2CO_3 vi) B_2H_6 vii) N_2O_4 viii) H_3PO_4 ix) Fe_2O_3 x) C_2H_2

Answer : Empirical formula is a simplest whole number ratio of atoms in the molecule, therefore the empirical formula of given compounds are: i) CH ii) CH_2

Space for Keynotes

BOARD QUESTION

1. A compound contains 34.8% oxygen, 52.2% carbon and 13.0% hydrogen. What is the molecular formula of the compound?

[Given that the molecular mass of the compound is 46g/mol]

Answer :

Element	Percentage	ATOMIC MASS	SIMPLEST RATIO	NUMBER OF ATOMS
Hydrogen	13.00g	$13.00/1 = 13.00$	$13.00/2.175 = 5.977 = 6$	6
Carbon	52.20g	$52.20/12 = 4.35$	$4.35/2.175 = 2$	2
Oxygen	34.80g	$34.80/16 = 2.175$	$2.175/2.175 = 1$	1

Empirical formula of the compound will be C_2H_6O , Empirical formula mass = 46 n = Molecular formula mass / Empirical formula mass = $46/46 = 1$, $\therefore$ Molecular formula = $(C_2H_6O) \times 1 = C_2H_6O$

LIMITING REAGENT

- Limiting reagent may be defined as the reactant which is completely consumed during the reaction. The reactant which is not completely consumed is referred to as excess reactant.
- The amount of the product obtained is determined by the amount of the reactant that is completely consumed in the reaction.

STOICHIOMETRIC CALCULATIONS

- The term Stoichiometry is derived from Greek words Stoicheion meaning element and metron meaning measure.
- In order to solve the problems based on chemical calculations the following steps are quite helpful.

> Write the balanced chemical equation. > From the chemical equation write the molar ratio between the given and required substance. > Convert these moles into the desired parameters such as mass or volume. > Calculate the final result by applying unitary method.

BOARD QUESTION

1. Calculate the mass of CaO obtained by heating of 2.5g CaCO_3 :

Answer : $\text{CaCO}_3(\text{s}) \rightarrow \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$ Number of moles of $\text{CaCO}_3 = (\text{Mass of CaCO}_3)/(\text{Mol.Mass of CaCO}_3)$
 $= 2.5/100 = 0.025$ Mole ratio of CaCO_3 and $\text{CaO}=1:1$ (from equation) Number of moles of $\text{CaO} = 0.025(1/1)=0.025$
 Mass of CaO formed $= 0.025 \times 56 = 1.4$ g.

BOARD QUESTION

1. In a reaction

$\text{A} + \text{B}_2 \rightarrow \text{AB}_2$ Identify the limiting reagent, if any, in the following reaction mixtures. i) 300 atoms of A + 200 molecules of B ii) 2 mol A + 3 mol B iii) 100 atoms of A + 100 molecules of B iv) 5 mol A + 2.5 mol B v) 2.5 mol A + 5 mol B Answer : i) According to the reaction, 1 atom of A reacts with 1 molecule of B. $\therefore$ 200 molecules of B will react with 200 atoms of A, thereby leaving 100 atoms of A unreacted. Hence, B is the limiting reagent. ii) According to the reaction, 1 mol of A reacts with 1 mol of B. $\therefore$ 2 mol of A will react with only 2 mol of B leaving 1 mol of B. Hence, A is the limiting reagent. (iii) 1 atom of A combines with 1 molecule of B. $\therefore$ All 100 atoms of A will combine with all 100 molecules of B. Hence, the mixture is stoichiometric and there is no limiting reagent. (iv) 1 mol of atom A combines with 1 mol of molecule B. $\therefore$ 2.5 mol of B will combine with only 2.5 mol of A. and 2.5 mol of A will be left unreacted. Hence, B is the limiting reagent. (v) 1 mol of atom A combines with 1 mol of molecule B. $\therefore$ 2.5 mol of A will combine with only 2.5 mol of B and the remaining 2.5 mol of B will be left. Hence, A is the limiting reagent

BOARD QUESTION

1. Calculate the mass of sulphur required to produce 11.2 litres of sulphur dioxide at NTP:

Answer : At NTP. 1 mole of any gas occupies 22.4 litres. Moles of $\text{SO}_2 = (\text{Volume of SO}_2)/(\text{Volume of 1 mole at NTP})$

- Moles of $\text{SO}_2 = 11.2/22.4$
- Moles of $\text{SO}_2 = 0.5$ moles

From the balanced equation $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$. 1 mole of S produces 1 mole of SO_2 . Therefore, moles of S = moles of SO_2

- Moles of S = 0.5 moles
- Mass of S = moles of S $\times$ molar mass of S
- Mass of S = 0.5×32
- Mass of S = 16 g

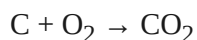
The mass of sulphur required to produce 11.2 litres of sulphur dioxide at STP is 16 g.

Space for Keynotes

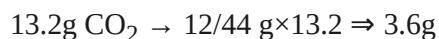
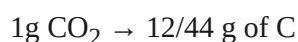
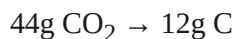
BOARD QUESTION

06. Calculate the mass of graphite that must be burnt to produce 13.2 g of CO₂ :

Answer :



$$12 + 32 = 44\text{g}$$

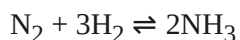


BOARD QUESTION

07. 50.0 kg of N₂ (g) and 10.0 kg of H₂ (g) are mixed to produce NH₃ (g). Calculate the mass of NH₃ (g) formed. Identify the limiting reagent in the production of NH₃ in this situation:

Answer :

The balanced chemical reaction is



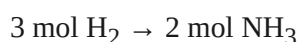
From the above balanced equation, 1 mol of N_{2(g)} reacts with 3 moles of H_{2(g)}.

$$\therefore \text{Moles of N}_2 = 50,000 \text{ g} / 28 \text{ g mol}^{-1} = 1785.71 \text{ mol}$$

$$\therefore \text{Moles of H}_2 = 10,000 \text{ g} / 2.106 \text{ g mol}^{-1} = 4960.3 \text{ mol}$$

$\therefore$ 1785.7 mol of N₂ react with $3 \times 1785.7 \text{ mol} = 5357.1 \text{ mol}$ but we have 4960.3 mol of H₂ only. Therefore, H₂ is a limiting reagent.

For the mass of NH₃ formed



$$\text{Moles of NH}_3 = 2/3 \times 5000 = 3333.3 \text{ mol}$$

$$\text{Mass of NH}_3 = 3333.3 \times 17 = 56666.7\text{g} = 56.7\text{kg}$$

TRY YOURSELF

TRY YOURSELF

1. The molecular mass of a hydrocarbon is 91 and its % composition is 92.4%C and 7.6%H. The molecular formula of compound depicts that n value is _____

TRY YOURSELF

TRY YOURSELF

1. Calculate the molecular formula of an organic compound containing 80% carbon and 20% hydrogen by mass (molar mass =30):

TRY YOURSELF

TRY YOURSELF

1. An organic compound has the following percentage composition C=12.36%;H=2.13%;Br=85%. Its molecular mass is 188. Find its molecular formula.
-

DAY 07

THE CONCENTRATION OF SOLUTIONS

- A solution is a homogeneous mixture of two or more components.
- Concentration of a solution refers to the amount of solute present in a given quantity of solution or solvent.
- Concentration of a solution can be expressed in the following ways.
 1. Mass Percentage
 2. Mole fraction
 3. Molarity
 4. Molality

MASS PERCENTAGE (% MASS OR W/W)

- Mass percentage may be defined as the number of parts by mass of solute per hundred parts by mass of the solution.
- A 5% aqueous solution of sugar by mass means that 100 g of solution contains 5 g of sugar or 5 g sugar is present in 95 g of water.
- Temperature independent.

Mass % of a component = (Mass of the component in the solution / Total mass of the solution) $\times$ 100
 Mass percentage(% by mass) of solute = $(W_B / W_A + W_B) \times 100$
 Mass percentage(% by mass) of solvent = $(W_A / W_A + W_B) \times 100$
 Sum of Mass percentages = 100

MOLE FRACTION [X]

- Mole fraction may be defined as the ratio of the number of moles of one component to the total number of moles of all components present in the solution.
- Mole fraction of solvent,

$$\chi_A = (n_A / n_A + n_B) = (w_A/M_A / w_A/M_A + w_B/M_B)$$

- Mole fraction of solute,

$$\chi_B = (n_B / n_A + n_B) = (w_B/M_B / w_A/M_A + w_B/M_B) \quad \chi_A + \chi_B = 1$$

- If equal no. of moles of solute and solvent are present ($n_A = n_B$), then

$$\chi_A = \chi_B = 1/2$$

BOARD QUESTION

1. Calculate the mass percentage of benzene (C_6H_6) and carbon tetrachloride (CCl_4) when 22 g of benzene is dissolved in 122 g of carbon tetrachloride?

Answer : Mass of solution = Mass of benzene + Mass of carbon tetrachloride = 22 g + 122 g = 144 g
 Mass percentage of benzene = (Mass of benzene / Mass of solution) $\times$ 100 = (22 g / 144 g) $\times$ 100 = 15.28%
 Mass percentage of carbon tetrachloride = (Mass of carbon tetrachloride / Mass of solution) = (122 g / 144 g) $\times$ 100 = 84.72%

MOLARITY [M]

- If equal weights of solute and solvent are present; then $W_A = W_B$

$$\text{Mole fraction of solvent, } \chi_A = \frac{1/M_A}{1/M_A + 1/M_B} \quad \text{Mole fraction of solute, } \chi_B = \frac{1/M_B}{1/M_A + 1/M_B}$$

Temperature independent

BOARD QUESTION

- Calculate the mole fraction of benzene in solution containing 30% benzene by mass in carbon tetrachloride?

Answer : Mass of benzene in 30% solution = 30 g Mass of carbon tetrachloride = (100-30)g = 70g Molar mass of benzene $C_6H_6 = (12 \times 6) + (1 \times 6) = 78$ g mol⁻¹ Molar mass of carbon tetrachloride, $CCl_4 = 12 + (35.5 \times 4) = 154$ g mol⁻¹ Moles of benzene in 30 g = $\frac{30}{78} = 0.385$ mol Moles of carbon tetrachloride in 70 g = $\frac{70}{154} = 0.455$ mol

$$\text{Total no. of moles} = 0.385 + 0.455 = 0.840 \quad \text{Mole fraction of benzene} = \frac{\text{No. of moles of benzene}}{\text{Total number of moles}} = \frac{0.385}{0.840} = 0.458$$

- Molarity of a solution is defined as the number of gram moles of the solute present in one litre of the solution or milli moles of solute present in one mL of solution.

$$\text{Molarity (M)} = \frac{\text{No. Moles of the solute}}{\text{Volume of solution in L}} \quad M = \frac{n_B}{V(L)} \quad M = \frac{n_B}{V(\text{mL})} \times 1000 \quad M = \frac{W_B(g)}{M_B} \times V(L) \quad M = \frac{W_B(g)}{M_B} \times V(\text{mL}) \times 1000$$

- Number of moles = Molarity $\times$ volume in L
- A solution having molarity equal to unity is called 1 molar solution i.e.; solution contains one mole of solute per litre of solution.

$$1M = 1\text{molar} = 1 \text{ mol L}^{-1}$$

- 5M aqueous solution of glucose means 5 moles of glucose are present in 1 L of solution
- Temperature dependent**

0.5 M	0.1 M	0.01 M
Semimolar solution	Decimolar solution	Centimolar solution

BOARD QUESTION

- Calculate the molarity of a solution containing 5 g of NaOH in 450 mL of solution?

$$\text{Answer : Number of moles} = \frac{\text{Given mass (g)}}{\text{Molar mass (g/mol)}} = \frac{5\text{g}}{40 \text{ g/mol}} = 0.125 \text{ moles} \quad \text{Volume in liters} = \frac{450 \text{ mL}}{1000} = 0.450 \text{ L}$$

$$\text{Molarity (M)} = \frac{\text{Number of moles}}{\text{Volume of solution in L}} \quad \text{Molarity (M)} = \frac{0.125 \text{ moles}}{0.450 \text{ L}} \approx 0.2778M \therefore \text{Molarity} \approx 0.28M$$

MOLALITY

BOARD QUESTION

1. Calculate the molarity of each of the following solutions:
2. 30 g of $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ in 4.3 L of solution
3. 30 mL of 0.5 M H_2SO_4 diluted to 500 mL.

Answer : Molarity is given by: $\text{Molarity} = \frac{\text{Moles of solute}}{\text{Volume of solution in litre}}$

$$(1) \text{ Molar mass of } \text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O} = 59 + 2(14 + (3 \times 16)) + (6 \times 18) = 291 \text{ g mol}^{-1} \text{ Moles of } \text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O} = \frac{30}{291} \text{ mol}$$

$$= 0.103 \text{ mol} \text{ Therefore, molarity} = \frac{0.103 \text{ mol}}{4.3 \text{ L}} = 0.023 \text{ M}$$

$$(2) \text{ Number of moles present in 1000 mL of } 0.5 \text{ M } \text{H}_2\text{SO}_4 = 0.5 \text{ mol} . \text{ Number of moles present in 30 mL of } 0.5 \text{ M } \text{H}_2\text{SO}_4 = \frac{0.5 \times 30}{1000} \text{ mol} = 0.015 \text{ mol} \text{ Therefore, molarity} = \frac{0.015}{0.5 \text{ L}} \text{ mol} = 0.03 \text{ M}$$

MOLALITY (m)

- Molality of a solution may be defined as the number of moles of the solute present in 1000 g (1 kg) of the solvent.

$$\text{Molality (m)} = \frac{\text{No. of moles of solute}}{\text{Mass of the solvent in g}} \times 1000 \text{ Molality (m)} = \frac{n_B}{W_A} (\text{kg}) \text{ m} = \frac{W_B}{M_B} \times W_A (\text{kg}) \text{ m} = \frac{W_B \times 1000}{M_B} \times W_A (\text{g})$$

- A solution containing one mole of solute per 1000 g of solvent has molality equal to one and is called a molal solution.

$$1 \text{ m} = 1 \text{ molal} = 1 \text{ mol Kg}^{-1}$$

- 5 m aqueous solution of glucose means 5 moles of glucose are present in 1 kg of solvent.
- Temperature independent

BOARD QUESTION

1. Calculate molality of 2.5 g of ethanoic acid (CH_3COOH) in 75 g of benzene.

$$\text{Answer : } m = \frac{\text{moles of } \text{CH}_3\text{COOH}}{\text{mass of benzene in kg}} \text{ moles of } \text{CH}_3\text{COOH} = \frac{2.5}{60} = 0.0417 \text{ mol Mass of benzene} = 75 \text{ g} = 0.075 \text{ kg}$$

$$m = \frac{0.0417}{0.075} = 0.556 \text{ mol kg}^{-1}$$

Space for Keynotes

BOARD QUESTION

1. Calculate the mass of urea (NH_2CONH_2) required to prepare 2.5 kg of 0.25 m aqueous solution.

Answer : No. of moles of urea = 0.25 moles Mass of solvent (water) = 1 kg = 1000 g Molar mass of urea = $14 + 2 + 12 + 16 + 14 + 2 = 60 \text{ g mol}^{-1}$ 0.25 mole of urea = $(60 \times 0.25) \text{ mole} = 15 \text{ g}$ Total mass of the solution = $1000 + 15 \text{ g} = 1015 \text{ g} = 1.015 \text{ kg}$ Thus, 1.015 kg of solution contain urea = 15 g 2.5 kg of solution will require urea = $(15/100) \times 2.5 \text{ kg} = 37 \text{ g}$

TRY YOURSELF

1. Calculate the mass percentage of glucose and water if 30 g of glucose is dissolved in 90 g of water?

TRY YOURSELF

1. What is the concentration of sugar ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) in mol L^{-1} if its 20 g are dissolved in enough water to make a final volume up to 2 L ?

TRY YOURSELF

1. What is the molarity of a solution containing 10g of NaOH 500 mL of solution?
-

STRUCTURE OF ATOM

DAY 01

- The existence of atoms has been proposed since the time of early Indian and Greek philosophers (400 B.C.)
- The word 'atom' has been derived from the Greek word 'a-tomio' which means 'uncuttable' or 'indivisible'.

DISCOVERY OF ELECTRON - CATHODE RAYS

- More information about electrons is obtained from the study of cathode rays, which were discovered by Julius Plucker during the experiments with gas discharge tubes.

To Vacuum pump

Cathode – Anode

high voltage

- A discharge tube is long glass tube, fitted with metal electrodes on either end across which high voltage can be applied.
- The tube is also connected to a vacuum pump for controlling the pressure of the gas inside the discharge tube.
- If the gas pressure is reduced to 10^{-4} atmospheres and sufficiently high voltage is applied across the electrodes, the side near anode of the tube glows with a faint greenish light.
- Further investigations revealed that it was caused due to the bombardment of the walls of the tube by rays travelling from cathode.
- These rays were called cathode rays because they started from cathode and ended at anode.

PROPERTIES OF CATHODE RAYS

- The cathode rays start from cathode and move towards the anode.
 - The cathode rays are not visible but their behaviour can be observed with the help of fluorescent or phosphorescent materials.
 - In the absence of electrical or magnetic field, cathode rays travel in straight lines.
 - In the presence of electrical or magnetic field, the behaviour of cathode rays are similar to that expected from negatively charged particles, suggesting that the cathode rays consist of negatively charged particles, called electrons.
 - The characteristics of cathode rays do not depend upon the material of electrodes and the nature of the gas present in the cathode ray tube.
-

CHARGE TO MASS RATIO OF ELECTRON²

- J.J. Thomson measured the ratio of electrical charge (e) to the mass of electron (m_e) by using cathode ray tube and applying electrical and magnetic field perpendicular to each other as well as to the path of electrons.

[Figure] Cathode ray tube diagram showing Anode, Cathode, Fluorescent Screen, and Magnet

- Amount of deviation of the particles from their path in the presence of electrical or magnetic field depends upon:

> The magnitude of the negative charge on the particle. > Greater the magnitude of the charge on the particle, greater is the interaction with the electric or magnetic field and thus greater is the deflection. > The mass of the particle. > Lighter the particle, greater the deflection. > The strength of the electrical or magnetic field. > The deflection of electrons from its original path increases with the increase in the voltage across the electrodes, or the strength of the magnetic field.

The value of $e/m_e = 1.758820 \times 10^{11} \text{ Ckg}^{-1}$

CHARGE ON THE ELECTRON

- The charge (e) on an electron was determined by R.A. Millikan in 1909 by 'oil drop' experiment
- In this method a spray of oil droplets is produced by an atomizer. The oil droplets enter the apparatus through a small hole and are allowed to fall in between two charged plates.
- The motion of the droplets is observed with a telescope.

[Figure] Oil drop experiment apparatus showing Charged Plate (+), Charged Plate (-), Oil Droplets, Atomizer, Telescope, Small Hole, X-ray from the source, and Oil droplet under observation

- The space between charged plates is irradiated with X-rays. The X-rays ionize the molecules of air.
- One or more electrons, thus produced may be absorbed by oil droplets. The oil droplet as a result becomes negatively charged.
- By measuring the velocity of a given oil droplet as it falls freely under influence of gravity and then in an electric field, it is possible to calculate charge (q) on the droplet.
- Millikan found charge on all oil droplets could be expressed as whole number multiple of e , which was considered to be electronic charge.
- The charge on the electron to be $-1.6 \times 10^{-19} \text{ C}$.
- The present accepted value of electrical charge is $-1.602176 \times 10^{-19} \text{ C}$.
- Mass of electron = $e / e/m_e$
- Mass of electron = $9.11 \times 10^{-31} \text{ kg}$

- Cathode ray is composed of electrons. β -rays are also electrons.

KEEP IT IN MIND

- Television picture tubes are cathode ray tubes and television pictures result due to fluorescence on the television screen coated with certain fluorescent or phosphorescent materials.
-

BOARD QUESTION

01. (i) Calculate the number of electrons which will together weigh one gram. (ii) Calculate the mass and charge of one mole of electrons.

Answer : (i) $\therefore$ Mass of one electron = 9.11×10^{-31} kg $\therefore$ Number of electrons weighing together one gram (10^{-3} kg) will be = $\frac{1}{(9.11 \times 10^{-31}) \times 10^{-3}} = 1.098 \times 10^{27}$ electrons

(ii) $\therefore$ Mass of one electron = 9.1×10^{-31} kg $\therefore$ Mass of one mole of electrons = $(9.1 \times 10^{-31}) \times (6.022 \times 10^{23}) = 5.486 \times 10^{-7}$ kg $\therefore$ Charge on one electron = 1.602×10^{-19} C $\therefore$ Charge on one mole of electrons = $(1.602 \times 10^{-19}) \times (6.022 \times 10^{23}) = 1.6 \times 10^4$ C = 9.65×10^4 C

BOARD QUESTION

03. In Milikan's experiment, static electric charge on the oil drops has been obtained by shining X - rays. If the static electric charge on the oil drop is -1.282×10^{-18} C, calculate the number of electrons present in it.

Answer : Given, charge = -1.282×10^{-18} C Charge carried by one electron = -1.6×10^{-19} C $\therefore$ No. of electrons present in it = $\frac{-1.282 \times 10^{-18}}{-1.6 \times 10^{-19}} = 8$

HOMEWORK

TRY YOURSELF

01. Calculate the number of electrons which will together weigh one kilogram.

BOARD QUESTION

02. A certain particle carries 2.5×10^{-16} C of static electric charge. Calculate the number of electrons present in it.

Answer : Given, Charge = 2.5×10^{-16} C Charge carried by one electron = -1.6×10^{-19} C $\therefore$ No. of electrons present in it = $\frac{2.5 \times 10^{-16} \text{C}}{1.6 \times 10^{-19} \text{C}} = 1560$

DAY 02

DISCOVERY OF PROTONS – ANODE RAYS

- Goldstein discovered the existence of a new type of rays in the discharge tube.
- He used a perforated cathode in the discharge tube.
- On passing the electric discharge at low pressure he observed a new type of rays streaming behind the cathode. These rays were named anode rays or canal rays.

[Figure] Discharge tube diagram showing perforated cathode, anode, high voltage generator, vacuum pump, and red glow at low pressure

PROPERTIES OF ANODE RAYS

- Anode rays travel in straight lines.
- Anode rays consist of material particles.
- Anode rays are deflected by electric field towards negatively charged plate. This indicates that they are positively charged.
- When a magnetic field is applied, anode rays get deflected in the direction expected for positive particles. It indicates that they are positively charged.
- Charge to mass ratio of the particles in the anode rays depends upon nature of the gas taken in the discharge tube.
- e/m ratio is maximum when hydrogen gas is taken in the discharge tube. This indicates that positive ions formed from hydrogen are lightest.
- These lightest positively charged particles were named protons.

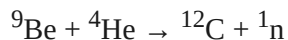
TRY YOURSELF

1. Calculate the mass and charge of five mole of electrons.

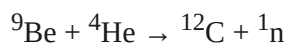
Space For Keynotes

DISCOVERY OF NEUTRONS

- Neutrons were discovered by James Chadwick.
- He bombarded a thin foil of beryllium with fast moving α -particles and observed that highly penetrating rays consisting of neutral particles were produced.



- These neutral particles were found to have mass 1.675×10^{-27} kg and were named neutrons.



Particle	Symbol	Absolute charge (C)	Mass (Kg)	Mass (u)
Electron	e	-1.6022×10^{-19}	9.11×10^{-31}	0.00054
Proton	p	$+1.6022 \times 10^{-19}$	1.67262×10^{-27}	1.00727
Neutron	n	0	1.67493×10^{-27}	1.00867

- It failed to explain the observations of Rutherford's α - particle scattering experiments.

RADIOACTIVITY

- Henri Becquerel observed that there are certain elements which emit radiation on their own and named this phenomenon as radioactivity and the elements known as radioactive elements.

[Figure] Diagram showing alpha, beta, and gamma rays being emitted from a source and their deflection in a magnetic field (alpha deflects one way, beta the opposite, gamma undeflected)

- It was observed that three kinds of rays i.e., α , β - and γ -rays are emitted.
- α -rays consists of high energy particles carrying two units of positive charge and four unit of atomic mass. That is, α -particles are identical to helium nuclei.
- β -rays are negatively charged particles identical to electrons.
- The γ - rays are high energy radiations like X-rays, are neutral in nature and do not consist of particles.
- As regards penetrating power, α -particles are the least, followed by β -rays (100 times that of α -particles) and γ -rays (1000 times of that α -particles).

THOMSON MODEL OF ATOM

[Figure] Diagram of Thomson's plum pudding model showing a uniform positive sphere with electrons embedded in it

- According to J. J. Thomson "an atom consists of a uniform sphere of positive electricity in which the electrons are distributed more or less uniformly".

- This model of atom is known as the 'plum-pudding model or raisin pudding model or watermelon model'.
 - This model assumed that mass of the atom is spread over the entire atom.
 - The model could explain electrical neutrality of atom.
-

KNOW YOUR HEROES

Henri Becquerel (1852–1908) was a French physicist who discovered radioactivity in 1896 while studying uranium salts. His work led to major advances in nuclear science. He shared the 1903 Nobel Prize in Physics with Marie and Pierre Curie.

RUTHERFORD'S ALPHA PARTICLE EXPERIMENT

- Rutherford and his students (Hans Geiger and Ernest Marsden) bombarded very thin gold foil with α -particles.
- A stream of high energy α -particles from a radioactive source was directed at a thin foil (thickness ~ 100 nm) of gold metal.
- The thin gold foil had a circular fluorescent zinc sulphide screen around it. Whenever α -particles struck the screen, a tiny flash of light was produced at that point.

[Figure] Diagram of Rutherford's alpha particle experiment setup showing source of α -rays, thin gold foil, ZnS screen, and circular fluorescent screen

RUTHERFORD'S OBSERVATIONS

- Rutherford observed that:

> Most of the α -particles (nearly 99%) passed through the gold foil undeflected. > Some of the α -particles were deflected by small angles. > A very few α -particles (1 in 20000) were either deflected by very large angles or were actually deflected back along their path.

[Figure] Diagram showing α particles being deflected by a nucleus (positive charge) in Rutherford's experiment

CONCLUSIONS

- Since most of the α -particles pass through the foil undeflected, it indicates that most of the space in an atom is empty.
- α -particles being positively charged and having considerable mass could be deflected only by some heavy, positively charged centre. The small angle of deflection of α -particles indicated the presence of a heavy positive centre in the atom.
- Rutherford named this positive centre as nucleus.
- α -particles which make head-on collision with heavy positive centre are deflected through large angles. Since the number of such α -particles is very small, the space occupied by the heavy positive centre must be very small.

HOMEWORK

TRY YOURSELF

1. Write the postulates of raisin pudding model of atom. Explain its failure ?

TRY YOURSELF

1. What are the conclusions of Rutherford's α - particle scattering experiment ?

TRY YOURSELF

1. Compare the properties of α , β and γ -rays ?

Space For Keynotes

DAY 03

RUTHERFORD'S NUCLEAR MODEL OF ATOM

- Most of the mass and all the positive charge of an atom is concentrated in a very small region called nucleus. Size of the nucleus is extremely small as compared with the size of the atom. Radius of the nucleus is of the order of 10^{-15} m, whereas radius of atoms is of the order of 10^{-10} m.
- The positive charge on the nucleus is due to protons. The magnitude of the charge on the nucleus is different for atoms of different elements.
- The nucleus is surrounded by electrons which are revolving around it at very high speeds. The electrostatic force of attraction between electrons and the nucleus is balanced by the centrifugal force acting on the revolving electrons.
- Total negative charge on the electrons is equal to the total positive charge on the nucleus so that atom on the whole is electrically neutral.
- Most of the space inside an atom is empty.

DRAWBACKS OF RUTHERFORD MODEL

- Rutherford's model failed to explain stability of atoms.
- it does not explain the distribution of electrons around the nucleus and does not tell us anything about their energies.
- Rutherford's model also failed to explain the existence of certain definite lines in the hydrogen spectrum.

ATOMIC NUMBER

- The number of unit positive charges on the nucleus of an atom of the element is called atomic number of the element
- In an atom, number of protons is equal to the number of electrons.
- Atomic number of an element is equal to the number of protons in the nucleus of its atom or the number of extra nuclear electrons.

MASS NUMBER

- Total number of protons and neutrons in the nucleus is called mass number of the atom.
- It is generally represented by the letter A.

Mass Number (A) = Number of protons + Number of neutrons = Number of nucleons

Representation of an Element

Mass number → A Symbol of the Atomic number → Z X → element

BOARD QUESTION

1. How many neutrons and protons are there in the following nuclei?

^{13}C , ^{16}O , ^{24}Mg , ^{56}Fe , ^{88}Sr Answer : Number of protons and neutrons in the following nuclei are as follows: ^{13}C ; Number of protons = 6 Number of neutrons = $13 - 6 = 7$ ^{16}O : Number of protons = 8 Number of neutrons = $16 - 8$

$_{12}^{24}\text{Mg}$; Number of protons =12 Number of neutrons =24-12=12

$_{26}^{56}\text{Fe}$; Number of protons =26 Number of neutrons =56-26=30

$_{38}^{88}\text{Sr}$; Number of protons =38 Number of neutrons =88-38=50

BOARD QUESTION

1. Find out the atomic number, mass number, number of protons, electrons and neutrons present in $_{8}^{16}\text{O}^{2-}$.

Answer : Atomic number (Z)=8 Mass number (A)=16 Number of protons = Atomic number (Z) $\therefore$ Number of protons =8 $\therefore$ Two negative charge means gain of two electrons Number of electrons =10. Number of neutrons = Mass number - Atomic number = A-Z = 16-8 = 8

ISOTOPES

- They are atoms of a given element which have the same atomic number but differ in their mass number.
- Eg: $_{1}^{1}\text{H}$, $_{1}^{2}\text{H}$ and $_{1}^{3}\text{H}$ are isotopes of hydrogen.

ISOBARS

- Isobars are the atoms of different element which have the same mass number but different atomic number.
- They have different number of electrons, protons & neutrons but sum of number of neutrons & protons remains same
- Eg: $_{6}^{14}\text{C}$ and $_{7}^{14}\text{N}$ are isobars

ISOTONES

- Isotones may be defined as the atoms of different elements containing same number of neutrons.
- Eg: $_{6}^{14}\text{C}$ and $_{8}^{16}\text{O}$ are isotones

ISOELECTRONIC SPECIES

- They are the atoms, molecules or ions which have the same number of electrons
- Eg: N^{3-} , O^{2-} , F^{-} , Ne , Na^{+} , Mg^{2+} , Al^{3+} are isoelectronic with each other

DEVELOPMENTS LEADING TO THE BOHR'S MODEL OF ATOM

MAXWELL'S WAVE THEORY

- Energy is emitted continuously from any source in the form of radiations travelling in the form of waves and are associated with electric and magnetic fields perpendicular to each other and to the direction of propagation.
- These waves are called electromagnetic waves or electromagnetic radiations.

[Figure] Graph showing electric field component and magnetic field component of electromagnetic waves with x, y, z axes labeled; direction of propagation indicated

- The electric and magnetic field components

of an electromagnetic wave have the same wavelength, frequency, speed and amplitude, but they vibrate in two mutually perpendicular planes.

PROPERTIES OF ELECTROMAGNETIC WAVES

- Electric and magnetic field oscillate perpendicular to each other.
- Both oscillate perpendicular to the direction of propagation of wave.
- Do not require any medium for propagation.
- Can travel in vacuum.
- Propagate at a constant speed that is with the speed of light.

TRY YOURSELF

1. How many nucleons are present in an atom of Nobelium, $^{254}_{102}\text{No}$? Calculate the number of electrons and neutrons also.

TRY YOURSELF

1. Calculate the number of electrons, protons and neutrons in each of the following species

i) H_3PO_4 ii) NH_4^+ iii) ClO_3^-

Mass numbers : $\text{H}=1$, $\text{P}=31$ $\text{Cl}=37$, $\text{O}=16$

Atomic numbers : $\text{H}=1$, $\text{P}=15$, $\text{Cl}=17$, $\text{O}=8$

Space For Keynotes

DAY 04

CHARACTERISTICS OF ELECTROMAGNETIC WAVES

WAVELENGTH (λ)

- It may be defined as the distance between two neighbouring crests or troughs of the wave.

$$1\text{\AA} = 10^{-10} \text{ m}, 1\text{nm} = 10^{-9} \text{ m}, 1\text{pm} = 10^{-12} \text{ m}$$

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FREQUENCY (ν)

- It is the number of waves which pass through a particular point in one second

$$1 \text{ cps} = 1 \text{ Hz}, 1 \text{ kHz} = 10^3 \text{ Hz}, 1 \text{ MHz} = 10^6 \text{ Hz}$$

VELOCITY (C)

- The distance travelled by a wave in one second is called velocity of the wave.
- Velocity of all electromagnetic radiations in space or in vacuum is same and is equal to $3 \times 10^8 \text{ m/sec}$

$$v = c/\lambda$$

WAVENUMBER ($\bar{\nu}$)

- The number of waves or wavelengths per unit length.

$$\bar{\nu} = 1/\lambda$$

AMPLITUDE

- It is the height of a crest or depth of a trough of a wave.

TIME PERIOD

- It is the time taken to complete one oscillation
- Time period = $1/\text{frequency}$

[Figure] Wave diagram showing amplitude, wavelength, crest, trough, and reference line with labels A, B, C, D, E

ELECTROMAGNETIC SPECTRUM

- It is an arrangement of various forms of electromagnetic radiations in order of increasing or decreasing wavelengths or frequencies.

[Figure] Electromagnetic spectrum diagram showing different regions (radio waves, microwaves, infrared, visible light, ultraviolet, X-rays, gamma rays) with wavelength ranges

PLANCK'S QUANTUM THEORY OF RADIATION

KEEP IT IN MIND

- The following observations could not be explained with the help of the electromagnetic theory.
- The nature of emission of radiation from hot bodies (black -body radiation).
- Ejection of electrons from metal surface when radiation strikes it (photoelectric effect).
- Variation of heat capacity of solids as a function of temperature.
- Line spectra of atoms with special reference to hydrogen.

BOARD QUESTION

1. Yellow light emitted from a sodium lamp has a wavelength (λ) of 580 nm . Calculate the frequency (ν) and wavenumber ($\tilde{\nu}$) of the yellow light.

Answer : Given, $\lambda = 580 \text{ nm} = 580 \times 10^{-9} \text{ m}$; $c = 3.0 \times 10^8 \text{ ms}^{-1}$ $\therefore$ Frequency $\nu = c/\lambda \therefore \nu = (3 \times 10^8 \text{ ms}^{-1})/(580 \times 10^{-9} \text{ m}) = 5.17 \times 10^{14} \text{ s}^{-1}$ $\therefore$ Wavenumber $\tilde{\nu} = 1/\lambda$

BOARD QUESTION

02. Calculate the wavelength, frequency and wavenumber of a light wave whose period is $2.0 \times 10^{-10} \text{ s}$.

Answer : Given, Time = $2.0 \times 10^{-10} \text{ s}$ Frequency (ν) = $1/T = 1/(2.0 \times 10^{-10} \text{ s}) = 5 \times 10^9 \text{ s}^{-1}$ Wavelength (λ) = $c/\nu = (3 \times 10^8 \text{ ms}^{-1})/(5 \times 10^9 \text{ s}^{-1}) = 6.0 \times 10^{-2} \text{ m}$ Wavenumber ($\tilde{\nu}$) = $1/\lambda = 1/(6 \times 10^{-2} \text{ m}) = 16.66 \text{ m}^{-1}$

- The Radiant energy is emitted or absorbed not continuously but discontinuously in the form of small packets of energy called quanta.
- The amount of energy associated with a quantum of radiation is proportional to the frequency of radiation.
- A body can emit or absorb energy only in terms of the integral multiples of quantum.

[Figure] Diagram showing photon packets with wavy lines and arrows, labeled "Photon" and " $E=h\nu$ "

ENERGY OF A PHOTON	TOTAL ENERGY OF RADIATION
$E = h\nu$	$E = nh\nu$
$E = hc/\lambda$	$E = nhc/\lambda$
$E = hc\tilde{\nu}$	$E = nhc\tilde{\nu}$

Space For Keynotes

HOMEWORK

TRY YOURSELF

1. The wavelength range of the visible spectrum extends from violet (400 nm) to red (750 nm).

Express these wavelengths in frequencies (Hz). (1 nm = 10^{-9} m)

TRY YOURSELF

1. Calculate
2. wavenumber and
3. frequency of yellow radiation having wavelength 5800 Å.

BOARD QUESTION

1. Find energy of each of the photons which

i) correspond to light of frequency 3×10^{15} Hz. ii) have wavelength of 0.50 Å.

Answer : i) Given, $\nu = 3 \times 10^{15}$ Hz $\therefore$ Energy of photon (E) = $h\nu \therefore E = 6.626 \times 10^{-34} \text{ Js} \times 3 \times 10^{15} \text{ s}^{-1} = 1.988 \times 10^{-18} \text{ J}$ ii) Given, $\lambda = 0.50 \text{ Å} = 0.50 \times 10^{-10} \text{ m}$ $E = h\nu = hc/\lambda$ $E = (6.626 \times 10^{-34} \text{ Js} \times 3 \times 10^8 \text{ m s}^{-1}) / (0.50 \times 10^{-10} \text{ m}) = 3.98 \times 10^{-15} \text{ J}$

BOARD QUESTION

1. What is the number of photons of light with a wavelength of 4000 pm that provide 1 J of energy?

Answer : Given, $\lambda = 4000 \text{ pm} = 4 \times 10^{-9} \text{ m}$ As we know that, $E = nh\nu$, $E = nhc/\lambda$, $n = E\lambda/hc$ $n = (1 \times 4 \times 10^{-9}) / (6.626 \times 10^{-34} \times 3 \times 10^8) = 2.012 \times 10^{16}$ photons

Space For Keynotes

DAY 05

PHOTOELECTRIC EFFECT

- The photoelectric effect was discovered in 1887 by the German physicist Heinrich Rudolf Hertz.
- Photoelectric effect may be defined as the phenomenon of ejection of electrons from the surface of a metal when light of suitable frequency strikes it.
- Radiations having a particular frequency called threshold frequency can only eject the electrons.
- The maximum kinetic energy of photoelectrons is directly proportional to the frequency of the incident radiation but is independent of its intensity.
- The number of photoelectrons emitted per second is directly proportional to the intensity of the incident radiation but does not depend upon its frequency.

[Figure] Diagram showing a photoelectric effect setup with a vacuum chamber, metal surface, detector, battery, and electrons being ejected from the metal surface

THRESHOLD FREQUENCY

- The threshold frequency is the lowest frequency below which photoelectric emission is impossible.
- The work function(ω_0) of a metal is related to threshold frequency as

$$\omega_0 = h\nu_0 \quad \omega_0 = hc/\lambda_0$$

KEEP IT IN MIND $h\nu = \omega_0 + KE$ $KE = h(\nu - \nu_0)$ $h\nu = h\nu_0 + \frac{1}{2}mv^2$ $KE = hc(1/\lambda - 1/\lambda_0)$ $h\nu = h\nu_0 + KE$ $KE = hc(\lambda_0 - \lambda)/(\lambda\lambda_0)$ - When $n < n_0$, Photoelectric effect doesnot occur. - When $n = n_0$, Photoelectric effect occurs and electrons are ejected with zero velocity. - When $n > n_0$, Photoelectric effect occurs and electrons are ejected with a characteristic velocity.

BOARD QUESTION

1. A photon of wavelength 4×10^{-7} m strikes on metal surface, the work function of the metal being 2.13 eV. Calculate i) the energy of the photon (eV) ii) the kinetic energy of the emission and iii) the velocity of the photoelectron ($1\text{eV} = 1.602 \times 10^{-19}$ J).

Answer : i) Energy of the photon $= h\nu = hc/\lambda = (6.63 \times 10^{-34} \text{Js} \times 3 \times 10^8 \text{ ms}^{-1}) / (4 \times 10^{-7} \text{ m}) = 4.97 \times 10^{-19} \text{ J}$ ii) Energy of photon = K.E. + Work function Kinetic energy = Energy of photon - Work function $= 4.97 \times 10^{-19} \text{ J} - 2.13 \times 1.602 \times 10^{-19} \text{ J} = 1.56 \times 10^{-19} \text{ J} = 0.97\text{eV}$ iii) $\frac{1}{2}mv^2 = 1.56 \times 10^{-19} \text{ J}$

$$v^2 = \frac{2 \times 1.56 \times 10^{-19}}{m} = \frac{2 \times 1.56 \times 10^{-19}}{9.1 \times 10^{-31}} \quad v^2 = 0.34 \times 10^{12} \quad v = 5.85 \times 10^5 \text{ ms}^{-1}$$

$$= \frac{6.63 \times 10^{-34} \text{ Js} \times 3 \times 10^8 \text{ ms}^{-1}}{6800 \times 10^{-10} \text{ m}} = 2.93 \times 10^{-19} \text{ J}$$

$$\text{Energy of photon} = \text{K.E.} + W_0 = 0 + W_0 = W_0 \therefore W_0 = 2.93 \times 10^{-19} \text{ J}$$

$$\text{Threshold frequency } v_0 = c / \lambda_0 = \frac{3 \times 10^8 \text{ ms}^{-1}}{6800 \times 10^{-10} \text{ m}} = 4.41 \times 10^{14} \text{ s}^{-1}$$

BOARD QUESTION

1. Electromagnetic radiation of wavelength 242 nm is just sufficient to ionise the sodium atom.

Calculate the ionisation energy of sodium in kJ mol^{-1} .

Answer : Energy of one photon of the radiation having wavelength 242 nm is just sufficient to ionise the sodium atom. $\therefore$ Ionisation energy of sodium = Energy of one photon of radiation of wavelength Energy of photon = $h\nu$

$$= \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \text{ Js} \times 3 \times 10^8 \text{ ms}^{-1}}{242 \times 10^{-9} \text{ m}} = 8.22 \times 10^{-19} \text{ J/ photon} = 8.22 \times 10^{-22} \text{ kJ/ photon} \therefore \text{Ionisation energy per mole} = 8.22 \times 10^{-22} \times 6.022 \times 10^{23} \text{ kJ/mol} = 495 \text{ kJ/mol}$$

BOARD QUESTION

1. Electrons are emitted with zero velocity from a metal surface when it is exposed to radiation of wavelength 6800 Å.

Calculate threshold frequency (v_0) and work function (W_0) of the metal.

$$\text{Answer : Energy of photon} = \frac{hc}{\lambda}$$

BLACK BODY RADIATION

- An ideal body which is a perfect absorber and perfect emitter of radiation is called black-body. The radiation emitted by such a body is called black-body radiation.
- When such a body is heated it becomes red hot. In other words, it emits red-coloured light. As the temperature is further increased, the colour of the radiation emitted changes from red to yellow to white and finally to purple as the temperature becomes very high.
- This implies that wavelength of maximum intensity of radiation emitted by blackbody decreases with an increase in temperature.

ATOMIC SPECTRA

- When an element is excited by some method such as by heating, by passing electric current or by passing electric discharge, the atoms of the element emit electromagnetic radiations of definite frequencies.
 - The arrangement of these radiations in the order of increasing wavelength or decreasing frequencies is called emission spectrum or atomic spectrum of the element.
 - Atomic spectra of most of the elements consist of a number of bright lines separated by dark bands. That is why atomic spectrum is also known as line spectrum.
-

- The element helium was discovered in the sun by study of spectrum of sunlight.
- Elements like rubidium (Rb), Caesium (Cs), indium (In), scandium (Sc), etc. were discovered when their minerals were analysed by spectroscopic methods.
- Isoelectronic species exhibit identical spectrum.

SPECTRUM → ABSORPTION SPECTRUM SPECTRUM → EMISSION SPECTRUM

The spectrum obtained consists of a large number of sharp lines. Each line corresponds to a particular frequency of light emitted by hydrogen atoms.

[Figure] Hydrogen spectrum showing energy level transitions for Lyman, Balmer, Paschen, Brackett, Pfund, and Humphrey series with $n=1$ to $n=8$ energy levels

ABSORPTION SPECTRUM

[Figure] Diagram showing gas in ground state absorbing radiation, with prism and screen displaying dark absorption lines

- The gas in ground state absorbs radiation of particular wavelengths and radiations of remaining wavelength passes through the gas which scatter through a prism and appears as bright lines on the screen.
- The dark lines show the missing wavelengths which are already absorbed by the gas and is no more available to pass through the prism.

EMISSION SPECTRUM OF HYDROGEN

- When an electric discharge is passed through hydrogen gas, the hydrogen molecules dissociate into hydrogen atoms. The excited hydrogen atoms emit electromagnetic radiation of discrete frequencies.

FOR ANY SERIES OF LINES

First Line	Last Line
• Minimum Energy	• Maximum Energy
• Maximum Wavelength	• Minimum Wavelength
• Minimum Frequency	• Maximum Frequency
• Minimum Wavenumber	• Maximum Wavenumber

First excited state, $n=2$ Second excited state, $n=3$ and so on

SERIES	n_1	n_2	REGION
Lyman series	$n_1 = 1$	$n_2 = 2, 3, 4, \dots$	Ultraviolet region
Balmer series	$n_1 = 2$	$n_2 = 3, 4, 5, \dots$	Visible region
Paschen series	$n_1 = 3$	$n_2 = 4, 5, 6, \dots$	Infrared region

SERIES	n_1	n_2	REGION
Brackett series	$n_1 = 4$	$n_2 = 5, 6, 7, \dots$	Infrared region
Pfund series	$n_1 = 5$	$n_2 = 6, 7, 8, \dots$	Infrared region
Humphrey's series	$n_1 = 6$	$n_2 = 7, 8, 9, \dots$	Far infrared region

RYDBERG FORMULA

• The wavelength of a spectral line can be calculated from its wavenumber by Rydberg Formula i.e., $\bar{\nu} = R \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$

n_1 and n_2 are the electronic levels involved in transition • R is a constant known as Rydberg constant. • $R = 109677 \text{ cm}^{-1}$ or $R = 13.59 \text{ eV} \approx 13.60 \text{ eV}$ ($1 \text{ eV} = 8.0655 \times 10^5 \text{ m}^{-1}$) • For hydrogen like species (one electron system) $\bar{\nu} = R \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] Z^2$ $\lambda = \frac{9.12 \times 10^{-6} Z^2}{\left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]} \text{ cm}$ $\nu = 3.29 \times 10^{15} \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \text{ s}^{-1}$ • If a line in the spectrum has $n_2 = \infty$, then that spectral line is called limiting line.

NUMBER OF SPECTRAL LINES

• When an electron jumps from n^{th} energy level to ground level (through all possible jumps), Number of spectral lines produced = $\frac{n(n-1)}{2}$ • When an electron jumps from n_2 energy level to n_1 energy level, Number of spectral lines produced = $\frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$

BOARD QUESTION

1. What is the wavelength of light emitted when the electron in a hydrogen atom undergoes transition from the energy level with $n = 4$ to an energy level with $n = 2$?

Answer : According to Rydberg's formula, $\bar{\nu} = 109,677 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] Z^2$

In the present case $n_2 = 4$ and $n_1 = 2$ $\bar{\nu} = 109,677 \left[\frac{1}{(2)^2} - \frac{1}{(4)^2} \right] Z^2$ $\lambda = \frac{1}{\bar{\nu}}$ $\bar{\nu} = \frac{1}{20564 \text{ cm}} = 4.86 \times 10^{-5} \text{ cm} = 4.86 \times 10^{-5} \times 10^7 \text{ nm} = 486 \text{ nm}$

BOARD QUESTION

1. Calculate the wavenumber for the longest wavelength transition in the Balmer series of atomic hydrogen.

Answer : The longest wavelength corresponds to minimum energy (ΔE) transition.

For Balmer series this transition is from $n_2 = 3$ to $n_1 = 2$. $\bar{\nu} = 109,677 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] Z^2$ $\bar{\nu} = 109,677 \left[\frac{1}{(2)^2} - \frac{1}{(3)^2} \right] Z^2$ $\bar{\nu} = 15233 \text{ cm}^{-1} = 1.5233 \times 10^4 \text{ cm}^{-1} = 1.5233 \times 10^6 \text{ m}^{-1}$

Space For Keynotes

HOMEWORK

BOARD QUESTION

1. Calculate the wavelength of the first and the last line in the Balmer series of hydrogen spectrum.

Answer : For the first line in Balmer series $n_1=2$ and $n_2=3$

$$\frac{1}{\lambda_f} = 109,677 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] = 109,677 \left[\frac{1}{2^2} - \frac{1}{3^2} \right]$$

$$= 109677 \times 5/36 \text{ cm}^{-1} = 15232.9 \text{ cm}^{-1}$$

$$\lambda_f = 6.565 \times 10^{-5} \text{ cm} = 656.5 \text{ nm}$$

For the last line in Balmer series $n_1=2$ and $n_2 = \infty$

$$\frac{1}{\lambda_f} = 109,677 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] = 109,677 \left[\frac{1}{2^2} - \frac{1}{\infty} \right]$$

$$= 109677 \times 1/4 \text{ cm}^{-1} = 27419.3 \text{ cm}^{-1}$$

TRY YOURSELF

1. If threshold wavelength (λ_0) for the ejection of an electron from metal is 330 nm, then calculate the work function for the photoelectric emission.

TRY YOURSELF

1. How many spectral lines are possible when electron in the 5th excited state jumps back to 2nd excited state ?

Space For Keynotes

DAY 06

BOHR MODEL

- Applicable only for single electron species like H, He^+ , Li^{2+} , Be^{3+} , B^{4+} etc.

POSTULATES

- The electron in the hydrogen atom revolves around the nucleus only in certain selected circular paths called orbits. These orbits are associated with definite energies and are also called energy shells or energy levels or stationary states. These are numbered as 1, 2, 3, 4, etc., are designated as K, L, M, N, etc. shells.

[Figure] Circular diagram showing concentric orbits labeled K shell, L shell, M shell, N shell around a nucleus

- Only those orbits are permitted in which the angular momentum of the electron is a whole number multiple of $h/2\pi$, where h is Planck's constant.

> Angular momentum of an electron in an atom is quantized.

$$mvr = n \frac{h}{2\pi}$$

where, m = mass of electron, v = velocity of electron, r = radius of orbit, n = number of orbit in which electron is present.

- As long as the electron remains in a particular orbit, it does not lose or gain energy. That is energy of an electron in a particular orbit re-

[Figure] Portrait of Niels Bohr

TRY YOURSELF

1. Calculate the wavenumber for the longest wavelength transition in the Paschen series of atomic hydrogen.

TRY YOURSELF

1. The wavenumber of the limiting line in the Lyman series of hydrogen is 109678 cm^{-1} . Calculate the wavenumber of the limiting line in the Balmer series of He^+ .

Main Chapter / Day Heading

Section Heading

Subsection Heading

Minor Heading

mains constant.

- Electron can jump from lower to higher orbit by absorbing energy in the form of photon

Energy ($h\nu$) absorbed

$$h\nu = \text{Energy Absorbed} = E_2 - E_1$$

- Electrons can jump from higher to lower orbit by releasing energy in the form of photon.

Energy ($h\nu$) emitted

$$\text{Energy Released} = E_2 - E_1$$

- The frequency of radiation absorbed or emitted when transition occurs between two stationary states that differ in energy by

$$\nu = \Delta E/h = (E_2 - E_1)/h$$

- Where E_1 and E_2 are the energies of the lower and higher allowed energy states respectively.
- This expression is known as Bohr's frequency rule.

RADIUS OF BOHR'S ORBIT

- For n^{th} orbit of any one electron species with atomic number Z

$$r_n = (0.529 \times n^2)/Z \text{ \AA} \quad r_n = r_0 \times n^2/Z = 0.529 n^2/Z$$

$$\text{i.e., } r \propto n^2$$

- Radius of the first orbit of Hydrogen-Bohr radius, $r_0 = 0.529 \text{ \AA}$

ENERGY

- The energy of a stationary state of hydrogen atom is given as $E_n = - (2\pi me^4)/(n^2 h^2)$ where n is the n^{th} energy level.
- Energy of n^{th} orbit of hydrogen or hydrogen-like one electron species is given as :

$$E_n = - (2\pi^2 me^4)/(n^2 h^2) Z^2$$

$$E_n = - 1312/n^2 Z^2 \text{ kJ/mol}$$

$$E_n = - 13.6/n^2 Z^2 \text{ eV/atom}$$

$$E_n = - 21.8 \times 10^{-19}/n^2 Z^2 \text{ J/atom}$$

ADVANTAGES OF BOHR MODEL

- It explains the stability of atom.
- Radius of various orbits, velocity of electron and energy of electron in various orbits of H and H- like species can be calculated.

- Line spectrum of hydrogen can be explained with Bohr's theory.
-

DRAWBACKS OF BOHR'S THEORY

- Bohr's model does not explain the spectrum of multielectron atom. It can only explain the spectrum of single electron species like H-atom, He^+ , Li^{2+} , Be^{3+} , etc.
- It does not explain the splitting of spectral lines under the influence of magnetic field (Zeeman effect) and in the electric field (Stark effect).
- Bohr's theory does not justify the principle of quantization of angular momentum.
- Bohr gave flat/two-dimensional model of atom by saying that electron revolves in circular orbit around nucleus in one plane. Whereas it is now established that atom has 3D structure.
- Bohr's Model fails to explain de-Broglie concept of dual character and Heisenberg's principle of uncertainty.

BOARD QUESTION 01. Applying Bohr's model calculate the wavelength of the radiation emitted when an electron in a hydrogen atom makes a transition from fourth energy level to first energy level. Planck's constant $h = 6.63 \times 10^{-34}$ Js.

Answer : Energy of fourth energy level, $E_4 = -\frac{2.18 \times 10^{-18}}{(4)^2}$ J/atom Energy of first energy level, $E_1 = -\frac{2.18 \times 10^{-18}}{(1)^2}$ J/atom $\Delta E = E_1 - E_4 = -2.18 \times 10^{-18} \left[\frac{1}{(1)^2} - \frac{1}{(4)^2} \right]$ J/atom $= -2.04 \times 10^{-18}$ J/atom. This energy is given out in the form of a single photon or quantum of radiation. The frequency of the photon (taking energy in terms of magnitude) is given by the expression $\therefore \Delta E = h\nu$ or $\lambda = \frac{hc}{\Delta E}$

$$(\therefore \nu = \frac{c}{\lambda})$$

Substituting for $h = 6.63 \times 10^{-34}$ Js and $c = 3.00 \times 10^8 \text{ m s}^{-1}$ We get $\lambda = \frac{6.63 \times 10^{-34} \text{ Js} \times 3.00 \times 10^8 \text{ m s}^{-1}}{2.04 \times 10^{-18} \text{ J}} = 9.7 \times 10^{-8} \text{ m} = 97 \text{ nm}$.

BOARD QUESTION 02. Applying Bohr's model, calculate the frequency and wavenumber of the radiation emitted when an electron in hydrogen atoms returns from fifth energy level to second energy level.

Answer : $\Delta E = E_2 - E_5 = \frac{-2.18 \times 10^{-18}}{(2)^2} \text{ J} - \frac{-2.18 \times 10^{-18}}{(5)^2} \text{ J} = 2.18 \times 10^{-18} \left[\frac{1}{(5)^2} - \frac{1}{(2)^2} \right] \text{ J} = -4.58 \times 10^{-19} \text{ J}$ ΔE is negative indicating that the energy is given out. The frequency of the photon emitted is given by the relation, $\Delta E = h\nu$ $\nu = \frac{\Delta E}{h}$

$$= \frac{4.58 \times 10^{-19} \text{ J}}{6.63 \times 10^{-34} \text{ Js}} = 6.91 \times 10^{14} \text{ s}^{-1} \quad \nu = \frac{v}{c} \quad v = \frac{c}{\nu} = \frac{3 \times 10^8 \text{ m s}^{-1}}{6.91 \times 10^{14} \text{ s}^{-1}} = 4.34 \times 10^{-7} \text{ m}$$

PATHWAY TO QUANTUM MECHANICAL MODEL

- Einstein suggested that light has dual nature i.e., particle nature as well as wave nature.
- De Broglie proposed that particle has dual nature, that is particle nature as well as wave nature.

DUAL NATURE OF MATTER

- Every particle of mass m moving with a velocity v is associated with certain wavelength λ .
- De-Broglie wavelength of material particles is given as :

$$\lambda = h/mv \quad \lambda = h/P$$

$$\lambda_1/\lambda_2 = P_2/P_1 = m_2v_2/m_1v_1$$

λ = wavelength, h = Planck's constant m = mass of particle v = velocity of particle $mv = p$ = momentum

HEISENBERG'S UNCERTAINTY PRINCIPLE

"It is impossible to determine exactly and simultaneously the position and velocity (momentum) of a sub atomic particle like electron at a given instant of time."

BOARD QUESTION

1. i) The energy associated with the first orbit in the hydrogen atom is $-2.18 \times 10^{-18} \text{ J atom}^{-1}$.

What is the energy associated with the fifth orbit? ii) Calculate the radius of Bohr's fifth orbit for hydrogen atom.

Answer : i) $E_n = -2.18 \times 10^{-18} / n^2 \text{ J} \therefore E_5 = -2.18 \times 10^{-18} / 5^2 = -8.72 \times 10^{-20} \text{ J}$ ii) For H-atom, $r_n = 0.529 \times n^2 \text{ \AA} \therefore r_5 = 0.529 \times 5^2 = 13.225 \text{ \AA} = 1.3225 \text{ nm}$.

BOARD QUESTION

1. Calculate the wavelength of a body of mass 1 mg moving with a velocity of 10 m sec^{-1} .

Answer : We know, $\lambda = h/mv$ $v = 10 \text{ m sec}^{-1}$ $h = 6.626 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}$ $\lambda = 6.626 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1} / 10^{-6} \text{ kg} \times 10 \text{ ms}^{-1} = 6.626 \times 10^{-29} \text{ m}$

BOARD QUESTION

1. Calculate the momentum of a moving particle which has a de Broglie wavelength of 200 pm.

Answer : $\lambda = h/mv$ or $mv = h/\lambda$ $h = 6.626 \times 10^{-34} \text{ Js} = 6.626 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}$ $\lambda = 200 \text{ pm} = 2 \times 10^{-10} \text{ m}$
Momentum, $mv = h/\lambda = 6.626 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1} / 2 \times 10^{-10} \text{ m} = 3.313 \times 10^{-24} \text{ kg m s}^{-1}$

- Ruled out the concept of definite orbits by the concept of probability.

$$\Delta x \cdot \Delta p \geq h/4\pi \quad \Delta x \cdot m\Delta v \geq h/4\pi \quad \Delta E \cdot \Delta t \geq h/4\pi$$

For numerical purpose we generally use.

$$\Delta x \cdot \Delta p = h/4\pi$$

$$\text{i.e. } \Delta x \cdot \Delta p = h/4\pi = 6.6 \times 10^{-34} / 4 \times 22/7 = 5.25 \times 10^{-35}$$

$$\Delta x \cdot m\Delta v = 5.25 \times 10^{-35}$$

- If the position of a particle is $\Delta x \rightarrow 0$ measured precisely. $\Delta p \rightarrow \infty$
- If the momentum of the particle $\Delta p \rightarrow 0$ is measured precisely. $\Delta x \rightarrow \infty$

$$\text{Answer : Mass of ball, } m = 150 \text{ g} = 150 \times 10^{-3} \text{ kg} = 0.150 \text{ kg}$$

$$\text{Uncertainty in position, } \Delta x = 1 \text{ \AA} = 10^{-10} \text{ m} \quad \Delta x \times m\Delta v = h/4\pi$$

$$\Delta v = h/4\pi \times \Delta x \times m = 6.626 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1} / 4 \times 3.14 \times 10^{-10} \text{ m} \times 0.150 \text{ kg} = 3.52 \times 10^{-24} \text{ m s}^{-1}$$

HOMEWORK

BOARD QUESTION

1. The uncertainty in the momentum of a particle is $6 \times 10^{-2} \text{ kg ms}^{-1}$.

Calculate the uncertainty in its position.

$$\text{Answer : Uncertainty in momentum, } \Delta p = 6 \times 10^{-2} \text{ kg m s}^{-1} \text{ According to Heisenberg's principle,}$$

$$\Delta x \times \Delta p = h/4\pi$$

$$\therefore \Delta x = h/\Delta p \times 4\pi$$

$$\Delta x = 6.6 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1} / 6 \times 10^{-2} \text{ kg m s}^{-1} \times 4 \times 3.14 = 8.76 \times 10^{-34} \text{ m}$$

BOARD QUESTION

1. Calculate the uncertainty in the velocity of a cricket ball of mass 150 g, if the uncertainty in its position is of the order of 1 Å ($h = 6.6 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}$)

TRY YOURSELF

1. The energy of an electron in the first orbit is -1312 kJ/mole. Calculate the energy of the electron in the second excited state
-

Try Yourself: 15. Calculate the energy required to excite an electron from $n = 1$ to $n = 3$ Try Yourself: 17. Calculate the energy associated with the first orbit of He^+ kJ/mol ? Try Yourself: 16. The energy of the electron of hydrogen atom in the ground state is -1312 kJ/mole . Calculate the energy of the electron of Li^{2+} ion in the ground state Try Yourself: 18. Calculate the wavelength of an electron moving with a velocity of $2.05 \times 10^7 \text{ m s}^{-1}$.

DAY 07

QUANTUM MECHANICAL MODEL OF ATOM

- This model is based on Schrodinger Wave Equation.
- For a system in which an atom or a molecule whose energy does not change with time, the Schrodinger equation can be written as

$$\hat{H}\psi = E\psi$$

Ψ AND $|\Psi|^2$

- A moving electron is associated with a wave and the wave function ψ gives the amplitude of electron wave. It has got no physical significance.
- The probability of finding an electron at a point within an atom is proportional to the square of the orbital wave function i.e., $|\psi|^2$ at that point. $|\psi|^2$ is known as probability density and is always positive.
- From the value of $|\psi|^2$ at different points within an atom, it is possible to predict the region around the nucleus where electron will most probably be found.

ORBITAL

- An orbital may be defined as that region of space around the nucleus where the probability of finding an electron is maximum (90-95%).

Orbit	Orbital
It is a well-defined circular path of electron around the nucleus.	It is a region of space around the nucleus where the electron is most likely to be found.

TRY YOURSELF

1. Calculate the uncertainty in position of an electron if the uncertainty in its velocity is $5.7 \times 10^5 \text{ ms}^{-1}$. ($h = 6.626 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}$; Mass of electron = $9.1 \times 10^{-31} \text{ kg}$)

TRY YOURSELF

1. What will be the mass of a particle if uncertainty in its position is 10^{-8} m and velocity is $5.26 \times 10^{-25} \text{ m s}^{-1}$?

It represents planar motion of an electron.	It represents three dimensional motion of an electron.
The maximum number of electrons in an orbit is $2n^2$.	An orbital can accommodate maximum of two electrons.
Orbits are circular in shape.	Orbitals have different shapes.
Orbits are non-directional in character hence they cannot explain shapes of molecules.	Orbitals (except s-orbitals) have directional character and hence they can account for shapes of molecules.
Against Heisenberg's principle	In accordance with Heisenberg's principle

QUANTUM NUMBERS

- In order to specify energy, size, shape and orientation of orbital three quantum numbers are required, which are derived from Schrodinger wave equation. These are principal quantum number, azimuthal quantum number and magnetic quantum number.
- In order to designate the electron, an additional quantum number called spin quantum number is needed to specify spin of the electron.

> Principal quantum number. > Azimuthal quantum number. > Magnetic quantum number. > Spin quantum number.

PRINCIPAL QUANTUM NUMBER (n)

- Determines the energy of an electron and the average distance of an electron from the nucleus(size).
- This quantum number tells us in which principal energy level or shell the electron is present.
- It can have any whole number value such as 1, 2, 3, 4,....., etc. The energy levels or energy shells corresponding to these numbers are designated as K, L, M, N,.....,etc.
- For hydrogen and hydrogen like species, the energy and size of the orbital are determined by principal quantum number alone.

AZIMUTHAL QUANTUM NUMBER (l)

- It determines angular momentum of the electron.
- The value of l gives the sub-level or subshell in which electron is located.
- It also determines the shape of the orbital in which the electron is located.
- It may have all possible whole number values from 0 to n-1 for each principal energy level.
- For a given value of n, l can have n values.
- Angular momentum, L is given as follows

$$L = \frac{h}{2\pi} \sqrt{l(l+1)}$$

- The various sub-levels are designated as s,p,d,f etc. depending upon the value of l, as follows:

Value of l	0	1	2	3	4	5
Designation of sub-shell	s	p	d	f	g	h

- When $n = 1$, $l = 0$ indicating 1s subshell
 - When $n = 2$, $l = 0, 1$ indicating 2s, 2p subshells
 - When $n = 3$, $l = 0, 1, 2$ indicating 3s, 3p, 3d subshells
 - When $n = 4$, $l = 0, 1, 2, 3$ indicating 4s, 4p, 4d, 4f subshells and so on.
-

MAGNETIC QUANTUM NUMBER (m_l) $m = 3$ digits

- It refers to the different orientations of electron cloud in a particular sub-shell. These different orientations are called orbitals.
- The possible values of m_l range from $+l$ through 0 to $-l$, thus making a total of $(2l+1)$ values.
- In a subshell, the number of orbitals is equal to $(2l + 1)$.

KEEP IT IN MIND An orbitals can accommodate maximum of 2 electrons.

Maximum number of electrons in a subshell = $2(2l + 1)$

Subshell	s	p	d	f	g	l value	0	1	2	3	4	Max. no. of electrons
	2	6	10	14	18							

SPIN QUANTUM NUMBER (s OR m_s)

- It denotes the spin motion of electron.
- It can have only two values which are $+1/2$ and $-1/2$
- The $+1/2$ value indicates clockwise spin (generally represented by an arrow pointing upwards i.e., $\uparrow$) and other indicates anti-clockwise spin (generally represented by an arrow pointing downwards i.e., $\downarrow$).
- Due to its spin the electron behaves as a tiny magnet.

Spin magnetic moment (μ) = $\sqrt{n(n+2)}$ B.M. n = number of unpaired electron

KEEP IT IN MIND - No. of subshells in n^{th} shell = n - No. of orbitals in n^{th} shell = n^2 - No. of electrons in n^{th} shell = $2n^2$ - No. of orbitals in a subshell = $2l + 1$ - No. of electrons in a subshell = $4l + 2$ - No. of radial nodes = $n-l-1$ - No. of angular nodes = l - No. of l values = No. of Subshells - No. of m values = No. of Orbitals - No. of s values = No. of electrons

BOARD QUESTION

1. Using s, p, d notations describe the orbitals with the following quantum numbers :

2. $n = 1, l = 0$ 2) $n = 3, l = 1$

3. $n = 4, l = 2$ 4) $n = 4, l = 3$.

Answer :

1. 1s 2) 3p 3) 4d 4) 4f

BOARD QUESTION

1. What are the possible values of l for an electron in (1) third energy level (2) 3d subshell ?

Answer :

1. For third energy level $n=3$. Since l may have values from 0 to $n-1$, the possible values of l are 0, 1 and 2.

2. For d sub-shell $l=2$, hence, in 3d-subshell the only possible value of l is 2.

BOARD QUESTION

1. How many sub-shells are there in N shell? How many orbitals are there in d sub-shell?

HOMEWORK

TRY YOURSELF

1. How many quantum numbers are needed to designate an orbital ? Name them.

TRY YOURSELF

1. The principal quantum number, n of an atomic orbital is 5 . What are the possible values of l ?

Answer :

For N shell principal quantum number, $n = 4$. Number of subshells in an energy level = n $\therefore$ Number of subshells in N shell = 4 For d subshell $l=2$ Number of orbitals in a subshell = $2l + 1$ Number of orbitals in a d subshell = $2 \times 2 + 1 = 5$.

BOARD QUESTION

1. Which of the following orbitals are not possible? 2d, 4f, 4g and 6d.

Answer :

2d is not possible. For d subshell $l=2$. When $n=2$, l can be 0 and 1 . Since l cannot be 2 for $n=2$ hence 2d subshell is not possible. 4f is possible. For f subshell $l=3$. When $n=4$, l can be equal to 3 . 4g is not possible. For g subshell $l=4$ When $n=4$, l cannot be equal to 4 . 6d is possible. For d subshell $l=2$. When $n=6$, l can be equal to 2 = $2 \times 2 + 1 = 5$.

Space For Keynotes

DAY 08

SHAPE OF ORBITALS

- An orbital is a region of space around the nucleus where the chances of finding an electron of certain energy are greatest (90-95 percent).
- Node: It is a region in an orbital where the probability of finding an electron is zero.

Type of Nodes Angular node/ Nodal plane Radial node/ Spherical node Number of nodal plane in an orbital = 1 Number of radial node in an orbital = $n-l-1$

[Figure] Plots of wave function and distance from the nucleus for 1s and 2s orbitals respectively

[[TRY YOURSELF]]

1. The azimuthal quantum number l of an orbital is 3 . What are the possible values of m ?

[[Space For Keynotes]]

p-ORBITAL ($l = 1$)

- p-orbitals are dumbbell in shape having two lobes
- For p-orbital $l = 1$. Hence, $m = -1, 0, +1$. Thus, p-orbital has three different orientations, designated as p_x , p_y and p_z
- The two lobes of p-orbitals are separated by a nodal plane.

Dumb bell shape of p-orbital

Shape of three p-orbitals

s-ORBITAL ($l = 0$)

- s-orbitals are spherical in shape
- There is vacant radial space between two successive s-orbitals is known as a radial node.

Shape of s-Orbital

Plots of the square of wave function and distance from the nucleus for 1s and 2s orbitals respectively

d-ORBITAL ($l = 2$)

- d-orbitals are double dumbbell in shape having four lobes
 - For d-orbitals, $l = 2$. Hence $m = -2, -1, 0, +1, +2$. Thus there are five d orbitals, designated as d_{xy} , d_{yz} , d_{zx} , $d_{x^2-y^2}$ and d_{z^2}
 - Each d-orbital has 2 nodal planes except d_{z^2}
-

ELECTRONIC CONFIGURATION

• The electronic configuration of atoms refers to the distribution of electrons in different orbitals of the atom. The following three principles determine the filling of orbitals in the ground state of an atom.

- Aufbau principle
- Hund's rule of maximum multiplicity
- Pauli's exclusion principle

AUFBAU PRINCIPLE

According to this principle, in the ground state, the atomic orbitals are filled in order of increasing energies such that the lowest energy orbital is filled first upto its maximum capacity then the next orbital starts and so on.

PAULI'S EXCLUSION PRINCIPLE

• "No two electrons in an atom can have the same set of all four quantum numbers". • "An orbital can accommodate maximum of two electrons with anti parallel spins".

[Figure] Double dumbbell shape of d-orbital with coordinate axes and five d-orbitals (d_{xz} , d_{z^2} , d_{yz} , d_{xy} , $d_{x^2-y^2}$) shown with their shapes

ENERGY OF ORBITAL

• For single electron (H-like) species, Energy of orbital $\propto$ value of 'n' • $1s < 2s = 2p < 3s = 3p = 3d < 4s = 4p = 4d = 4f$. • For multielectron species, energy of orbital $\propto$ value of 'n+1'. This is known as (n + 1) rule. If 'n + 1' value is same for two or more orbitals, then energy will be more for the orbital having large 'n' • Orbitals having the same energy are called degenerate orbitals. For example $2p_x$, $2p_y$ and $2p_z$ are degenerate orbitals

[Figure] Energy level diagram showing orbital filling order with arrows indicating increasing energy levels

[Figure] Portrait of Wolfgang Pauli with Nobel Prize in Physics (1945) caption and electron spin diagrams showing correct (paired anti-parallel spins) and incorrect (parallel spins) configurations

HUND'S RULE OF MAXIMUM MULTIPLICITY

ELECTRONIC CONFIGURATION

This rule is the foundation for the filling of degenerate orbitals in the same subshell. Electron filling will not occur in orbitals of the same energy until all available orbitals of a given subshell contains one electron each with parallel spin.

"Friedrich Hund"

Incorrect Correct Correct

Atomic Number	Symbol	Electron Configuration
1	H	1s ¹
2	He	1s ²
3	Li	[He]2s ¹
4	Be	[He]2s ²
5	B	[He]2s ² 2p ¹
6	C	[He]2s ² 2p ²
7	N	[He]2s ² 2p ³
8	O	[He]2s ² 2p ⁴
9	F	[He]2s ² 2p ⁵
10	Ne	[He]2s ² 2p ⁶
11	Na	[Ne]3s ¹
12	Mg	[Ne]3s ²
13	Al	[Ne]3s ² 3p ¹
14	Si	[Ne]3s ² 3p ²
15	P	[Ne]3s ² 3p ³
16	S	[Ne]3s ² 3p ⁴
17	Cl	[Ne]3s ² 3p ⁵
18	Ar	[Ne]3s ² 3p ⁶
19	K	[Ar]4s ¹

Atomic Number	Symbol	Electron Configuration
20	Ca	$[\text{Ar}]4s^2$
21	Sc	$[\text{Ar}]4s^2 3d^1$
22	Ti	$[\text{Ar}]4s^2 3d^2$
23	V	$[\text{Ar}]4s^2 3d^3$
24	Cr	$[\text{Ar}]4s^1 3d^5$
25	Mn	$[\text{Ar}]4s^2 3d^5$
26	Fe	$[\text{Ar}]4s^2 3d^6$
27	Co	$[\text{Ar}]4s^2 3d^7$
28	Ni	$[\text{Ar}]4s^2 3d^8$
29	Cu	$[\text{Ar}]4s^1 3d^{10}$
30	Zn	$[\text{Ar}]4s^2 3d^{10}$

KEEP IT IN MIND Cr and Cu has exceptional electronic configuration due to their half-filled and fully filled configuration

EXTRA STABILITY OF HALF – FILLED AND COMPLETELY FILLED ORBITALS SYMMETRY It is a well-known fact that symmetry leads to stability. Subshells that are entirely or partially filled have a symmetrical distribution of electrons and are thus, more stable. EXCHANGE ENERGY

► When two or more electrons with the same spin are present in degenerate orbitals, electrons have a tendency to swap locations, and the energy produced is known as exchange energy. When the subshell is half-filled or totally filled the maximum number of exchanges is possible. As a result, stability is maximum.

4 exchange by electron 1 3 exchange by electron 2

2 exchange by electron 3 1 exchange by electron 4

$\mu = \sqrt{n(n + 2)} \text{ BM}$ • Where, n = no. of unpaired electrons and Bohr magneton (BM) is the unit of the magnetic moment.

WRITING ELECTRONIC CONFIGURATION OF IONS

• In ionisation, electrons are removed from the outermost shell of the electronic configuration of atom.

Fe : $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$ Fe^{2+} : $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$ Fe^{3+} : $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$

Cr : $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$ Cr^{2+} : $1s^2 2s^2 2p^6 3s^2 3p^6 3d^4$

• By electronic configuration, we can find out whether the substance is paramagnetic/ diamagnetic, i.e, magnetic property. • Spin only magnetic moment can be calculated by the formula

BOARD QUESTION

1. An atom has 2K,8L,5M electrons. Write down its electronic configuration and indicate in it

i) number of sub-shells ii) number of orbitals iii) number of unpaired electrons iv) number of electrons having $l=1$.

Answer : In the given atom there are $(2+8+5)=15$ electrons. The electronic configuration of the atom is: $1s^2, 2s^2, 2p^6, 3s^2, 3p_x^1, 3p_y^1, 3p_z^1$ • Number of subshells is 5 • Number of orbitals is 9 • Number of unpaired electrons is 3 • Number of electrons having $l=1$ is 9 .

BOARD QUESTION

1. An atom of an element contains 29 electrons and 35 neutrons. Deduce (i) the number of protons and (ii) the electronic configuration of the element.

Answer : • Number of protons = Number of electrons =29 • Electronic configuration of the element is: $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^{10}, 4s^1$

BOARD QUESTION

1. Write electronic configurations of Fe^{2+} and Fe^{3+} ions. Which of these has more number of unpaired electrons? Atomic no. of Fe is 26.

Answer : Fe atom has 26 electrons. Fe^{2+} and Fe^{3+} ions are formed by removal of two and three electrons respectively from Fe atom and hence contain 24 and 23 electrons respectively. Their electronic configurations are : $\text{Fe}^{2+} : 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^6$ $\text{Fe}^{3+} : 1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^5$ Fe^{2+} has 4 unpaired electrons while Fe^{3+} has 5 unpaired electrons. Therefore, Fe^{3+} has more number of unpaired electrons.

TRY YOURSELF

1. Out of 4f and 6s sub-shells which has higher energy and why?

HOMEWORK

TRY YOURSELF

1. Why 3d sub-shell has higher energy than 4s subshell in a multi-electron atom?

TRY YOURSELF

1. Which of the following is the ground state electronic configuration of carbon ($Z=6$)?

i) $\text{C}: 1s^2, 2s^2, 2p_x^2$ ii) $\text{C}: 1s^2, 2s^1, 2p_x^1, 2p_y^1, 2p_z^1$ iii) $\text{C}: 1s^2, 2s^2, 2p_x^1, 2p_y^1$.
